

Cross-Modal Retrieval from Coarse-Grained to Fine-Grained Perspectives: A Survey

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Abstract Cross-modal retrieval (CMR) has become a fundamental technique in multimedia understanding and recommendation systems, enabling information retrieval across heterogeneous modalities such as images, videos, and text. While several prior surveys have reviewed the progress of CMR, they are limited by outdated taxonomies and insufficient coverage of recent developments. In particular, most surveys focus on coarse-grained retrieval, which retrieves entire instances given a query, while neglecting fine-grained tasks that require retrieving at a finer semantic level to distinguish different subcategories or retrieving only a specific part of the instance, such as a region within an image or a segment within a video. Moreover, due to the rapid development of large-scale vision-language pre-training (VLP) models and multimodal large language models (MLLMs), many existing surveys fail to capture the impact of these transformative advancements on CMR. To address these gaps, in this survey, we provide a unified taxonomy, categorizing CMR into coarse-grained cross-modal retrieval (CCMR) and fine-grained cross-modal retrieval (FCMR). CCMR aims to retrieve the whole instance based on the given query, such as image-text and video-text retrieval. FCMR aims to distinguish and retrieve specific subordinate-level fine-grained categories within a super-class, or retrieve a part of the instance, such as image grounding and video temporal grounding. Taking both these types into consideration brings a broad view to CMR, bridges the gap between disparate tasks, and offers a comprehensive overview of the field. We review major methodological paradigms, including recent VLP-based and MLLM-based approaches, and summarize widely used datasets and evaluation protocols. Beyond systematic performance comparisons, we also discuss applications and insights for future research.

Keywords cross-modal retrieval, image-text retrieval, video-text retrieval, image grounding, video temporal grounding

1 Introduction

In our increasingly connected world, multimodal data, such as visual and textual content, are growing exponentially. Performing associations among different modalities is essential to obtain a comprehensive understanding. Therefore, cross-modal retrieval (CMR) has been a fundamental topic in the community. CMR aims to retrieve target information, such

as images and videos, for a given query, where the target and query are from different modalities.

The core objective of CMR is to bridge the significant semantic gap between heterogeneous modalities. Early work focused on statistical correlation analysis^[1, 2], while later deep learning methods map modalities into a common space^[3]. However, a primary challenge remains: distinguishing the correct target from hard negatives (i.e., samples that are highly similar

but semantically incorrect). For instance, in coarse-grained retrieval, the model must differentiate between videos of the same protagonist engaged in different activities. In fine-grained scenarios, this challenge is even more acute, requiring the model to discriminate between objects of the same category within an image (e.g., a specific bird described in the text versus other birds present), subtle variations in activities within a video, or subordinate-level categories that exhibit high intra-class variance and low inter-class variance. This demands a nuanced understanding that goes beyond simple entity recognition.

To systematically analyze these challenges and diverse methods in the field, we categorize CMR based on the granularity of the retrieval target. This taxonomy naturally separates the problem into two main branches.

Coarse-Grained Cross-Modal Retrieval (CCMR). This is the classic instance-level retrieval task^[4-7]. CCMR aims to retrieve the whole instance (e.g., an entire image or video) that semantically corresponds to a given query.

Fine-Grained Cross-Modal Retrieval (FCMR). In contrast, FCMR advances cross-modal retrieval to a finer granularity. On the one hand, it involves retrieving multimodal content belonging to specific fine-grained categories, which requires distinguishing visually similar objects within the same superclass, such

as different species of birds and model types of cars. On the other hand, it also includes retrieving a specific part of an instance rather than a whole image or video, such as a region in an image or a temporal segment in a video^[8, 9]. This requires a much deeper and more fine-grained alignment. In essence, CCMR addresses the challenge of global semantic matching, while FCMR tackles the more complex problem of localized correspondence, often dealing with more severe hard-negative interference. Both are essential for a comprehensive understanding and interaction with multimodal data.

Image-text retrieval (ITR) and video-text retrieval (VTR) are typical CCMR tasks. As shown in Fig.1, ITR^[10] takes a text query as input and retrieves the corresponding image, while VTR refers to the retrieval between the modalities of text and video. FCMR pushes the retrieval boundaries to a more granular level. We discuss fine-grained retrieval from two perspectives. 1) *Subcategory-Level Retrieval*. Here, the retrieval unit remains the entire instance, as in CCMR, but the task requires distinguishing among fine-grained subcategories, where a subcategory is defined as a specific category within a superclass^[11] with high intra-class variance and low inter-class variance (e.g., different bird species, car model types). 2) *Instance-Part Retrieval*. In this case, the retrieval target is more granular than the whole instance, focus-

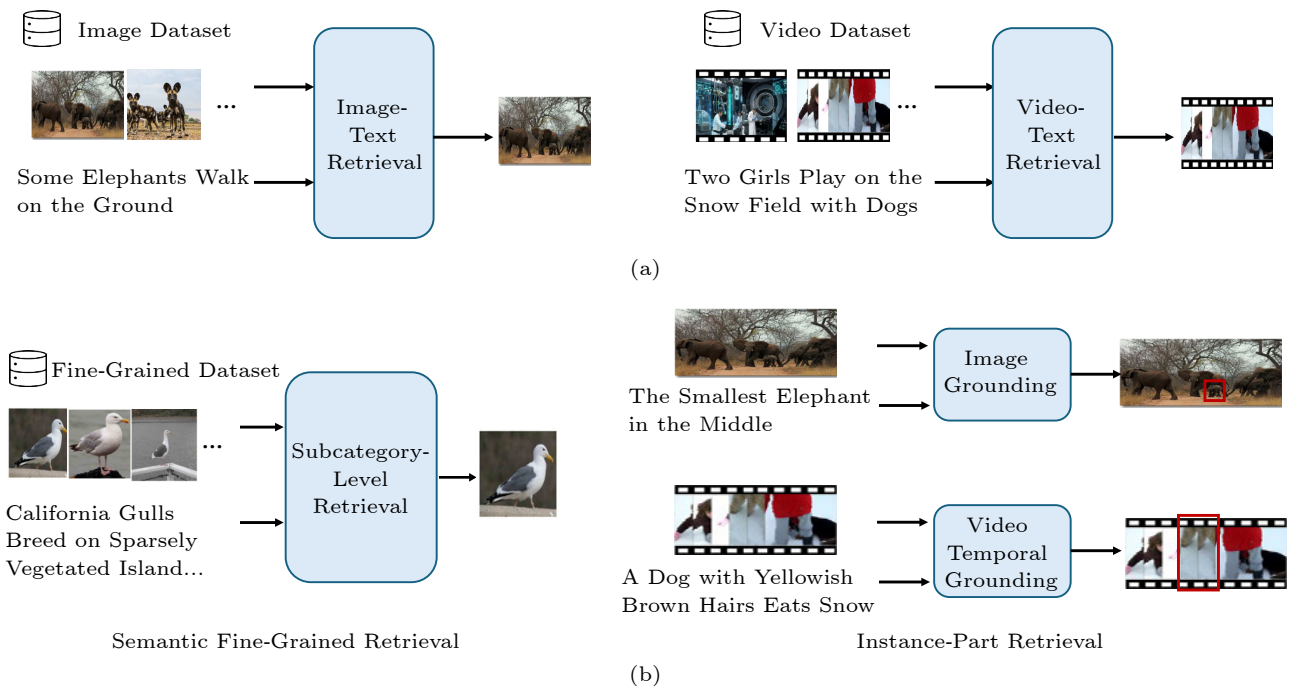


Fig.1. Illustration of coarse-grained and fine-grained cross-modal retrieval tasks. (a) Coarse-grained cross-modal retrieval. (b) Fine-grained cross-modal retrieval.

ing on specific parts such as spatial regions within images or temporal segments within videos. Two representative tasks are image grounding (IG) and video temporal grounding (VTG). As shown in Fig.1, image grounding aims to localize and retrieve certain image regions based on natural language references. Video temporal grounding, also known as video moment retrieval^[8, 9], aims to retrieve moments in a video given a user query.

Although several previous surveys have reviewed CMR, they have limitations in terms of timeliness, taxonomy, and comprehensiveness. Specifically, Wang *et al.*^[12] provided early insights into CMR, but their work did not include models proposed in recent years, thus necessitating a summary and classification of the new work. In recent years, vision-language pre-training models have developed rapidly, while the survey of Kaur *et al.*^[13] lacks classification and elaboration of such work. Cao *et al.*^[14] mainly summarized work on coarse-grained image-text retrieval, without covering fine-grained retrieval in diverse scenarios. Table 1 compares our survey with previous surveys on cross-modal retrieval. As we can see, we propose a new perspective from retrieval granularity, unify and introduce multiple tasks including retrieval and grounding, and discuss the applications of the latest vision-language pre-training (VLP) models and multimodal large language models (MLLMs) in cross-modal retrieval tasks.

This survey presents a comprehensive study on the recent progress of cross-modal retrieval. We adopt a broad view, from both coarse-grained and fine-grained perspectives, and extend the discussion of traditional cross-modal retrieval to a generalized one, which consists of CCMR and FCMR. As shown in Fig.2, our classification systematically reveals the deep intrinsic connections and differences across di-

verse retrieval tasks.

On the one hand, this taxonomy highlights the deep intrinsic connections between different tasks. 1) Both CCMR and FCMR address the core challenge of cross-modal alignment and are actively advancing across image, video, and other modalities. 2) Both CCMR and FCMR are empowered by the recent advanced large-scale VLP models. For example, the CLIP model provides a generalizable unified cross-modal representation and is widely used in both CCMR and FCMR. 3) From the perspective of task essence, both CCMR and FCMR aim to retrieve the most relevant target from a candidate set for a given query. Specifically, CCMR and subcategory-level FCMR target complete instances, while instance-part retrieval focuses on specific components. By integrating these tasks, this taxonomy provides a novel perspective on the multi-dimensional evolution of cross-modal retrieval from coarse-grained to fine-grained tasks. 4) Methodologically, the two are closely intertwined: CCMR increasingly incorporates fine-grained modeling (e.g., local alignment) to enhance global matching precision. Meanwhile, for FCMR methods, subcategory-level retrieval employs CCMR as a filter to narrow the search space, eliminating candidates not belonging to the target superclass. Instance-part retrieval, particularly two-stage approaches, effectively leverages CCMR progress by transforming fine-grained localization into a proposal-based coarse-grained retrieval problem.

On the other hand, this perspective also helps us understand the differences between CCMR and FCMR. 1) CCMR searches for complete instances (e.g., entire images or videos) from massive databases, whereas FCMR requires distinguishing visually similar subcategories or targeting specific parts within an instance (e.g., spatial regions or temporal segments).

Table 1. Comparison of Surveys on Cross-Modal Retrieval

Survey	Year	Task Coverage	Classification View	Granularity Focus	VLP Model	MLLM
Wang <i>et al.</i> ^[12]	2016	ITR, VTR	Methodologies	Coarse-grained	×	×
Kaur <i>et al.</i> ^[13]	2021	ITR, VTR	Methodologies	Coarse-grained	×	×
Cao <i>et al.</i> ^[14]	2022	ITR	Methodologies	Coarse-grained	✓	×
Wang <i>et al.</i> ^[15]	2025	ITR, VTR	Methodologies	Coarse-grained	✓	×
Xiao <i>et al.</i> ^[16]	2024	IG	Methodologies	Fine-grained	✓	✓
Zhang <i>et al.</i> ^[17]	2023	VTG	Methodologies	Fine-grained	✓	×
Wu <i>et al.</i> ^[18]	2025	VTG	Methodologies	Fine-grained	×	✓
Ours	2026	ITR, VTR, IG, VTG, subcategory-level fine-grained retrieval, compositional retrieval	Retrieval granularity, methodologies	Coarse & fine-grained	✓	✓

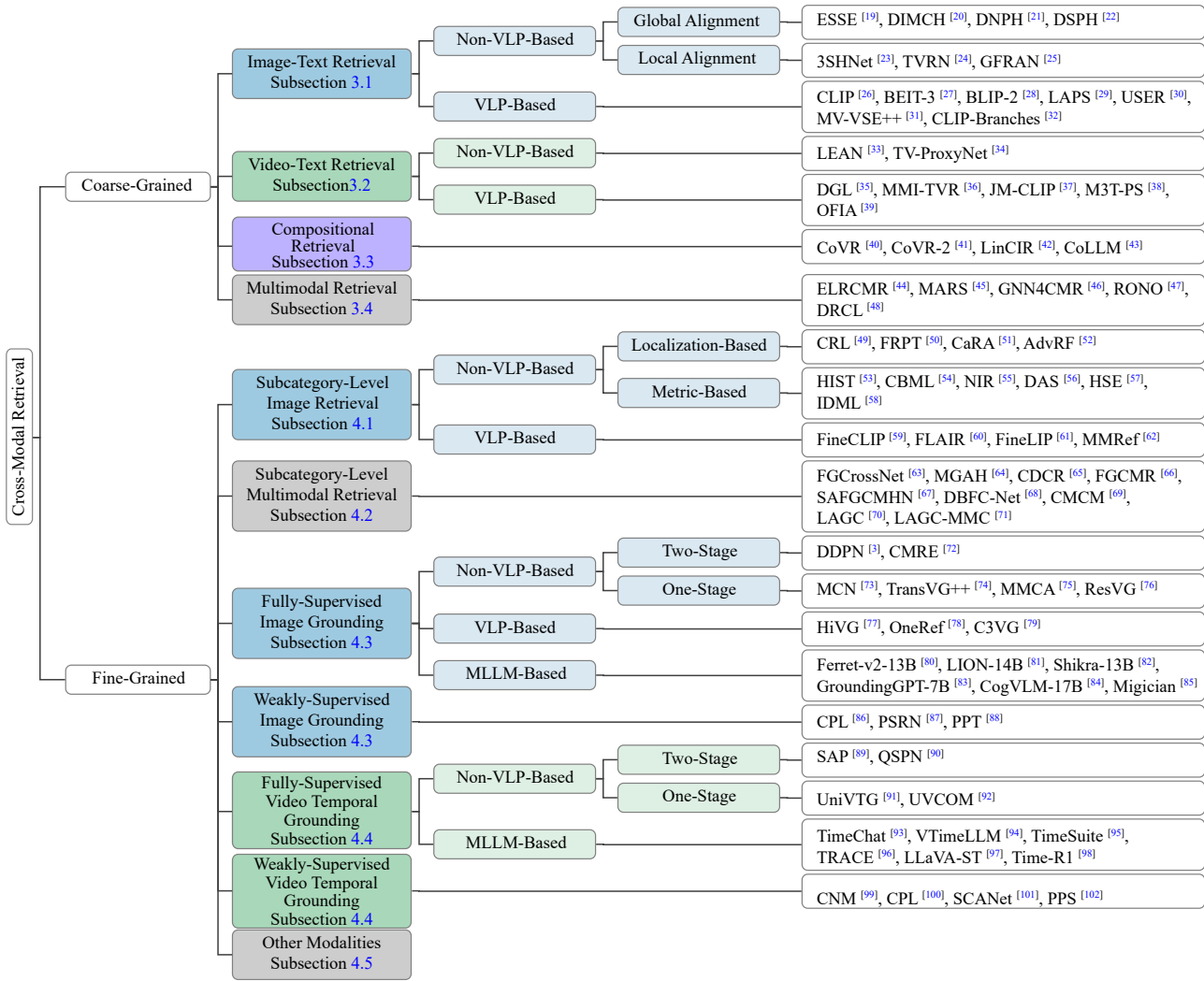


Fig.2. Overview of cross-modal retrieval tasks and their categorization.

2) This difference dictates distinct efficiency considerations and model architectures. Since CCMR requires retrieving from large-scale databases, it prioritizes efficiency^[10], leading to extensive research on Hash-based methods. In contrast, MLLMs, which typically employ computationally intensive architectures, are widely used in instance-part level FCMR^[103], where the search space is limited to a single instance, but are less prevalent in CCMR due to the prohibitive cost of applying such heavy fusion across massive candidates. Furthermore, although the search unit of subcategory-level retrieval is the entire instance, which is the same as CCMR, subcategory-level retrieval necessitates latent part-level cross-modal alignment similar to instance-part retrieval, because distinguishing fine-grained subcategories primarily depends on discriminative parts (e.g., tail feathers of a bird or headlights of a car). 3) Regarding supervision, this perspective clarifies the logic of weakly-supervised

fine-grained instance-part retrieval, which creatively utilizes coarse-grained CCMR annotations (e.g., video-level labels) to train systems for fine-grained tasks.

The rest of this paper is organized as follows. In [Section 2](#), we give an overview of recent work on CCMR and FCMR. Then the methods of CCMR and FCMR are detailed in [Section 3](#) and [Section 4](#), respectively. The experimental setting and performance comparison for all tasks mentioned above are presented in [Section 5](#). We further discuss practical applications and provide insights for future research directions in [Section 6](#) and [Section 7](#), respectively, offering a comprehensive perspective on the field.

2 Overview

In this section, we provide an overview of recent studies on CMR by categorizing them according to re-

trieval granularity. The overall framework is illustrated in Fig. 2. This classification is grounded in four key observations.

1) Overall, we categorize them into coarse-grained cross-modal retrieval (CCMR) and fine-grained cross-modal retrieval (FCMR) according to the retrieval granularity. In terms of objectives, both are information retrieval tasks; the distinction lies solely in the search space. Coarse-grained retrieval searches for complete images or videos from a massive database, whereas fine-grained retrieval searches for specific instances (objects or temporal segments) within a single image or video. Besides, both face the same core challenge of bridging the cross-modal semantic gap to achieve robust cross-modal alignment. Finally, both CCMR and FCMR involve text, image, video, and various other modalities. 2) Regarding task setup, weakly-supervised setup in fine-grained cross-modal retrieval involves training models using only coarse-grained annotations (image-text or video-text pairs), while aiming to achieve fine-grained retrieval capabilities (e.g., localizing specific regions or video segments) during inference. This setup greatly reduces annotation costs, as annotating object bounding boxes is typically time-consuming. 3) From the perspective of pre-trained model utilization, this taxonomy elucidates how foundation models impact different granularities. On the one hand, both categories extensively explore VLP models (e.g., CLIP) to enhance alignment. On the other hand, a divergence exists regarding MLLMs. Coarse-grained retrieval rarely employs MLLMs due to efficiency constraints; the “early fusion” architecture typical of MLLMs is computationally prohibitive when retrieving from large-scale databases, whereas it remains feasible for the limited search space of fine-grained tasks. 4) Regarding non-VLP-based methods, structural connections are evident in the retrieval stages. While coarse-grained retrieval is typically one-stage, some fine-grained retrieval methods involve two-stage approaches. The proposal generation phase in two-stage methods effectively transforms the fine-grained task into a coarse-grained retrieval problem by matching the query against a set of candidate proposals, further demonstrating the intrinsic link between these two categories.

2.1 Coarse-Grained Cross-Modal Retrieval

CCMR aims to retrieve entire instances. Its main

tasks include: image-text retrieval (ITR), video-text retrieval (VTR), compositional retrieval, and CCMR for other modalities, such as audio-text retrieval, image-audio retrieval, text-to-3D point cloud retrieval.

2.1.1 Image-Text Retrieval

The objective of the ITR task is to bridge the heterogeneous gap between images and text and explore their correlation. Existing methods tend to learn a semantic feature space, where an image is close to the matched text and away from other unmatched text. Recent studies can be grouped into non-VLP-based methods and VLP-based methods. Specifically, non-VLP-based methods^[104] mainly consist of global alignment methods and local alignment methods. VLP-based methods^[26] feature the paradigm of pre-training and fine-tuning, which obtains universal knowledge in the pre-training stage and specific domain knowledge in the fine-tuning stage. Benefiting from large-scale data, VLP-based methods have shown great advantages in many tasks.

2.1.2 Video-Text Retrieval

VTR aims at obtaining the target video from the dataset for a given sentence query, which requires an overall understanding of the video. Existing VTR methods can be divided into non-VLP-based methods and VLP-based methods. The former ones explore the feature extraction, the integration of multimodal features derived from various experts, and the alignment and interaction across different modalities. VLP-based methods leverage the knowledge of models pre-trained on large-scale external data. In VLP-based methods, recent CLIP-based methods have become popular in order to reduce computational cost.

2.1.3 Compositional Retrieval

The evolution of cross-modal retrieval has progressed to the more advanced compositional retrieval^[40]. Composed multimodal retrieval allows users to query visual data, such as images or videos, by providing a reference visual input combined with textual modifications. This approach aims to find target items that resemble the reference but are altered according to the user’s text-based feedback, offering enhanced flexibility and precision in search intent. Based on retrieval targets, the existing compositional

retrieval methods mainly include composed image retrieval and composed video retrieval.

2.1.4 Multimodal Retrieval

The application of CCMR has expanded beyond traditional image-text or video-text retrieval, extending into a generalized multimodal retrieval framework that encompasses diverse modalities such as audio, 3D data, and beyond^[105]. This expansion demonstrates the versatility of CCMR principles in establishing semantic associations across arbitrary modalities. However, multimodal retrieval introduces new challenges. As the diversity of modalities increases, the model needs to address heterogeneity among different feature spaces, noisy label effects inherent in diverse data sources, and scalability across varying modality combinations. For instance, audio-text and image-audio retrieval tackle the challenge of aligning descriptive text or static visual information with dynamic, temporal acoustic features. These frontier explorations are driving the development of modality-specific encoders and more universal cross-modal alignment technologies.

2.2 Fine-Grained Cross-Modal Retrieval

In contrast to CCMR, FCMR pushes the boundaries of retrieval to a more granular level. This fine-grained perspective on retrieval can be understood through two primary directions. The first is subcategory-level fine-grained retrieval, where the retrieval unit remains the entire instance, but the task demands distinguishing between subordinate-level categories (e.g., different bird species). The second direction involves retrieving a specific part of an instance, where the retrieval target is more granular than the whole. This includes tasks like image grounding, which performs retrieval of a spatial region within an image, and video temporal grounding, which focuses on the retrieval of a temporal segment within a video.

2.2.1 Subcategory-Level Image Retrieval

Subcategory-level image retrieval focuses on distinguishing subtle visual differences among subordinate fine-grained categories within the same superclass, such as specific bird species or car models. Methods are generally categorized into non-VLP-based and VLP-based methods. Non-VLP-based

methods rely on category supervision and can be further divided into localization-based approaches and metric-based approaches. The former explicitly highlights discriminative regions to suppress background noise, while the latter optimizes the similarity space. VLP-based methods leverage the zero-shot power of VLP models such as CLIP. Recent research either utilizes CLIP as a teacher model to distill fine-grained knowledge into student networks or directly enhances CLIP's capacity for local-level alignment and precise text modeling.

2.2.2 Subcategory-Level Multimodal Retrieval

Subcategory-level multimodal retrieval extends specific modality pairs to any-to-any retrieval among diverse modalities, including image, text, video, and audio, etc. It aims to establish a unified and discriminative space where subtle semantic correspondences among different fine-grained subcategories are preserved regardless of the input modality combination. Methods primarily focus on learning unified and discriminative multimodal representations. While early work utilized adversarial learning to bridge modality gaps, recent unified models employ joint constraints to learn shared embeddings. More advanced architectures further incorporate cross-modal attention mechanisms^[70] to capture fine-grained details, enhance feature interaction, and reduce distribution discrepancies. By breaking modality barriers, category-level FCMR retrieval provides a more holistic and flexible framework for FCMR.

2.2.3 Image Grounding

Image grounding is a typical instance-part retrieval task. It focuses on image region retrieval, aiming to localize and retrieve specific image regions guided by natural language references. The system must understand the semantics of the phrase and perform a fine-grained spatial search within the image to retrieve the corresponding region. Methods for this task can be broadly grouped into three categories: non-VLP-based methods, VLP-based methods, and MLLM-based methods. Non-VLP-based methods can be further divided into two-stage methods^[3], one-stage methods, and transformer-based methods^[74]. VLP-based methods leverage large-scale pre-trained vision-language models to learn robust shared embeddings. MLLM-based methods^[82, 83] utilize the powerful reasoning and generalization capabilities of MLLMs to

achieve state-of-the-art performance.

Besides, considering the huge labor of annotations, the weakly-supervised image grounding setting^[86] has been proposed, where only image-sentence annotations are provided. The weaker supervision provides less informative signals, making the task more challenging.

2.2.4 Video Temporal Grounding

Video temporal grounding, also known as video moment retrieval^[8, 9], extends the concept of instance-part retrieval to the temporal domain. The goal is to retrieve a specific temporal segment in a given untrimmed video for the given sentence query. Existing approaches can be categorized into non-VLP-based methods and MLLM-based methods. Non-VLP-based methods mainly include two-stage methods^[89] and one-stage methods. Recently, MLLM-based methods^[93, 96] have emerged, leveraging the temporal reasoning capabilities of LLMs to localize moments in videos, often achieving superior zero-shot performance.

Similar to weakly-supervised IG (image grounding), weakly-supervised video temporal grounding (VTG) was proposed to alleviate the issue of huge cost of complete annotations^[99–102].

2.2.5 FCMR for Other Modalities

Fine-grained cross-modal retrieval (FCMR) is not limited to image, text, or video modalities but also

extends to modalities such as audio and 3D data. Among them, 3D visual grounding^[106–109] is a popular task, aiming to align language queries with regions in 3D scenes. Recent advances with dedicated 3D feature extractors and encoders have improved the modeling of geometric and semantic relations, leading to more precise and robust retrieval. These developments broaden the applicability of FCMR and enable richer multimodal understanding in real-world scenarios.

3 Coarse-Grained Cross-Modal Retrieval

CCMR aims to retrieve coarse-grained targets, such as images in ITR and videos in VTR, typically by the similarity of feature representations.

3.1 Image-Text Retrieval

As shown in Fig.3, recent image-text retrieval methods can be grouped into non-VLP-based methods and VLP-based methods.

3.1.1 Non-VLP-Based Methods

Non-VLP-based methods consist mainly of global alignment methods and local alignment methods. These methods focus on learning effective feature representations and alignments, sometimes incorporating hash encoding strategies for efficiency.

1) Global Alignment

Global alignment based methods aim to directly learn the global representation of the whole image

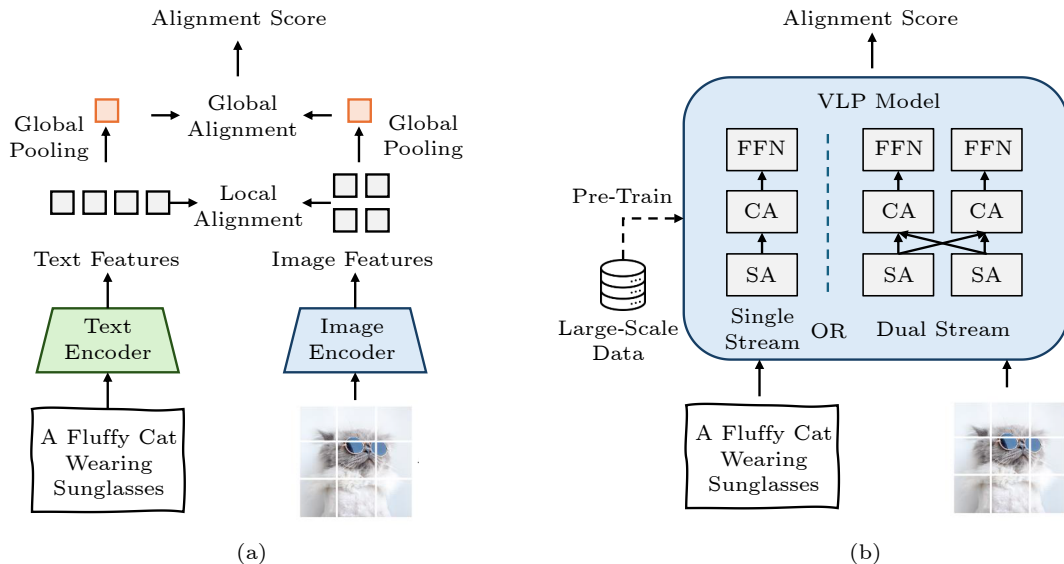


Fig.3. Illustration of two typical image-text retrieval methods. FFN, CA, and SA denote feed-forward network, cross-attention, and self-attention, respectively. (a) Non-VLP-based. (b) VLP-based.

and the description of the text. ESSE^[19] introduces sector embedding with bounded angular uncertainty to better capture one-to-many semantic correspondences while avoiding the limitations of unbounded variance. Global alignment methods are characterized by their simplicity and computational efficiency. They offer a fast and straightforward approach for instance-level matching, making them suitable for applications requiring rapid retrieval over large databases.

To obtain better efficiency and lower storage cost, hash encoding is also introduced in global alignment. DNPH^[21] uses learnable category proxies to solve the ambiguous neighborhood problem by enforcing intra-class compactness and inter-class separation. DSPH^[22] extends proxy-based hashing to multi-label retrieval by using a semantic-aware loss to jointly learn positive multi-label associations while penalizing irrelevant data-pair similarities.

2) Local Alignment

Local alignment involves methods that utilize the fine-grained correlation between image regions and text words. Besides, global alignment optionally acts as complementary global information to the local alignment involved methods. 3SHNet^[23] achieves high-precision and high-efficiency image-sentence retrieval by performing visual semantic-spatial self-highlighting without textual dependency, enabling fast inference. TVRN^[24] introduces a mutual refinement network that enhances image-text matching by first enriching textual features with visual context. Effective interactions inject cross-modal information to guide the final alignments. GFRAN^[25] introduces a fine-grained multimodal interaction and global-guided fragment aggregation mechanism to enhance image-text retrieval. Local alignment methods significantly improve retrieval accuracy by focusing on fine-grained correspondences between image regions and words. The primary drawback of this approach is its increased computational complexity, making it unsuitable for resource-limited scenarios.

Hash encoding strategies are also applied in local alignment to improve efficiency. DIMCH^[20] is a supervised, many-to-many alignment framework, which learns a set of diverse instance features for each modality.

3.1.2 VLP-Based Methods

Recently, cross-modal models featuring VLP have drawn great attention due to their remarkable suc-

cess in bridging the gap between visual and textual information. These models have the ability to learn rich and joint representations of images and text through pre-training on extensive datasets. VLP-based paradigm refers to the process where universal cross-modal models are pre-trained first using large-scale cross-modal data and then fine-tuned in the downstream tasks.

Leveraging the CLIP^[26] model, USER^[30] devises a new strategy to harness and transfer knowledge, which enriches the representation learning process for improved image-text retrieval. MV-VSE++^[31] introduces multiple embeddings and a unified multi-view loss, enabling better generalization across both traditional non-VLP-based and VLP-based models for image-text retrieval. CLIP-Branches^[32] addresses interactive text-image retrieval by introducing a fine-tuning phase using decision branches on CLIP embeddings to refine results based on user feedback.

VLP-based methods represent the state of the art, achieving dominant performance with remarkable zero-shot and transfer learning capabilities. Their improvements stem from two key aspects. 1) *Robust Generalization Capabilities*. Unlike traditional methods trained on closed-set data, VLP models leverage contrastive learning on massive image-text pairs to learn universal visual-semantic representations. This enables strong zero-shot and few-shot capabilities, allowing models to generalize well to Out-of-Distribution (OOD) data and unseen concepts without extensive retraining. 2) *Architectural Innovations*. In addition to the massive amounts of training data, the architectural advances in VLP models are also driving performance improvements. For example, recent architectures like ALBEF introduce the Align-before-Fuse strategy. By combining the efficiency of dual-encoders for global alignment with the precision of multimodal fusion encoders for fine-grained interaction, these architectures effectively bridge the modality gap and handle hard negatives better than relying solely on data scale. The main disadvantages are their prohibitive computational cost and data requirements.

3.1.3 Summary

In summary, the approaches to image-text retrieval offer different trade-offs. Non-VLP-based methods serve as a classic baseline, providing good model interpretability. VLP-based methods have achieved superior performance through large-scale

pre-training and represent the current dominant paradigm, but they come with high computational costs. In contrast, non-VLP-based methods incorporating hash encoding excel in retrieval speed and storage efficiency, making them more suitable for large-scale industrial applications with strict speed requirements.

3.2 Video-Text Retrieval

Video-text retrieval (VTR) is similar to image-text retrieval (ITR), except for the replacement of image modality with video modality. The final prediction is obtained by similarity ranking in the shared feature space as well. As shown in Fig.4, existing VTR methods can be categorized into two types: non-VLP-based methods and VLP-based methods. Non-VLP-based methods investigate better feature representation extraction, fusion of multimodal features obtained by multiple experts, and cross-modal alignment and interaction. VLP-based methods tend to utilize the knowledge of models pre-trained on large-scale data, such as the CLIP[26] model.

3.2.1 Non-VLP-Based Methods

Non-VLP-based methods usually include feature extraction and feature alignment. They primarily rely on task-specific architectures and training strategies to capture details necessary for distinguishing closely related videos. Feature representations of video, text, and optionally additional multimodal fea-

tures are first extracted from experts. LEAN[33] models high-order and n -ary relationships between semantic chunks for text-video retrieval by constructing a multimodal hypergraph. TV-ProxyNet[34] enhances text-video retrieval by decomposing the conventional one-to-many matching into proxy-based one-to-one alignments, thereby improving semantic precision.

Non-VLP-based methods enable the design of specialized and often complex fusion blocks to strengthen the interaction between video frames and text words, which can lead to better alignment for specific tasks.

3.2.2 VLP-Based Methods

Recently, numerous VLP-based methods have been proposed, especially those based on CLIP[26]. MMI-TVR[36] uses large language model (LLM) distillation to generate and align textual and visual knowledge skeletons for enhanced text-video retrieval. DGL[35] adapts VLP models by generating cross-modal prompts and capturing video context via global-local attention. JM-CLIP[37] extracts multi-grained features from text and videos to compute an extensive set of intra-modal and inter-modal similarities, which are then intelligently fused into a final score for contrastive learning using an attention mechanism. M3T-PS[38] accelerates text-to-video retrieval by predicting discrete semantic tags and leveraging an inverted index to efficiently filter candidate videos before fine-grained matching. While global contrastive learning has become the standard for video-text re-

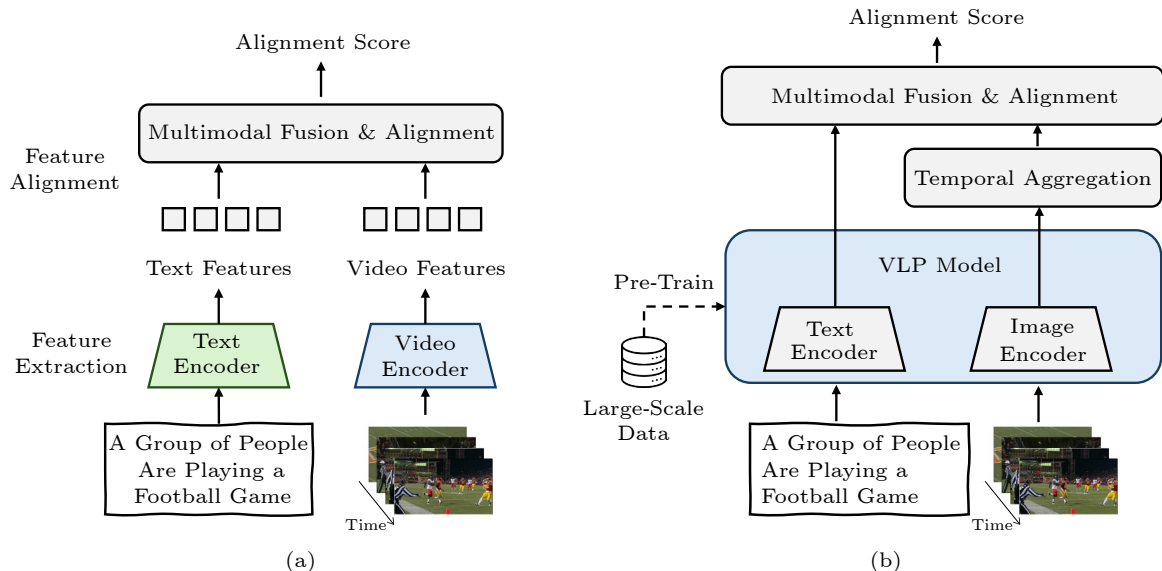


Fig.4. Illustration of two typical video-text retrieval methods. (a) Non-VLP-based. (b) VLP-based.

trieval, recent work aims to capture finer correspondences without expensive dense annotations. OFIA^[39] introduces an object-centric fine-grained alignment framework that utilizes pre-trained object detectors to extract region-level features from video frames. It utilizes an object extraction unit that encodes objects via cross-attention with the entire frame, preserving vital visual context during the fine-grained alignment process. While discriminative approaches based on contrastive learning have dominated text-video retrieval, they often overlook the underlying data distribution, leading to poor generalization.

VLP-based methods have become the dominant paradigm by successfully transferring knowledge from large-scale pre-trained models. This approach achieves state-of-the-art performance while being more resource-efficient than training a large video-language model from scratch. The primary disadvantages are inherited from the foundation models they are built on. The initial pre-training requires immense computational resources and massive datasets. While fine-tuning is more efficient than pre-training, it may also lead to a decrease in the pre-trained model's generalization ability due to catastrophic forgetting.

3.2.3 Summary

In summary, VLP-based methods, particularly those that utilize pre-trained models like CLIP, have become the mainstream paradigm. By transferring large-scale knowledge, they achieve state-of-the-art performance and are more efficient than training from scratch. However, the demand for large-scale data and substantial computational resources from these pre-trained large models also makes these non-VLP-based methods still important for vertical domains with limited computational resources or specific data constraints.

3.3 Compositional Retrieval

Composed multimodal retrieval allows users to query visual data, such as images or videos, by providing a reference visual input combined with textual modifications. Compositional image retrieval enables more nuanced and specific image searches by combining a reference image with a natural language description of desired changes. LinCIR^[42] proposes a language-only training framework using a self-masking projection (SMP) strategy to efficiently map text em-

beddings to the token space without image-text triplets. CoLLM^[43] leverages LLMs for composed query understanding and generates synthetic triplets from image-caption pairs to enhance training.

Composed video retrieval extends the compositional query concept to the temporal domain, allowing users to search for videos using a reference video or image along with text specifying modifications to actions, objects, or contexts. CoVR^[40] incorporates detailed language descriptions to enrich query-specific context and jointly learns vision-only, text-only, and vision-text embeddings. CoVR-2^[41] extends CoVR by introducing an automated pipeline to construct a large-scale dataset of 1.6 million triplets from web-scraped video-caption pairs to support composed video retrieval.

3.4 Multimodal Retrieval

The principles of coarse-grained cross-modal retrieval (CCMR) are not limited to image-text and video-text interactions but have been extended to a wide range of other modalities, including text, image, video, audio, etc. These extensions demonstrate the versatility of CCMR in handling diverse forms of multimodal data. However, the heterogeneity among different modalities presents significant challenges for multimodal retrieval, such as the alignment of heterogeneous feature spaces, the mitigation of noisy label effects, and the scalability to diverse modalities.

To address these challenges, various methods have been proposed, each targeting specific aspects of multimodal retrieval. ELRCMR^[44] further enhances the learning process by projecting multimodal data into a shared feature space using contrastive learning. It incorporates early learning regularization to prevent the memorization of noisy labels during training and adopts a dynamic weight balance strategy to alleviate clustering drift, ensuring stable and robust training. Moving beyond regularization techniques, MARS^[45] jointly trains LabNet and DataNet to preserve the semantic similarity between label-derived representations and modality-shared representations. This approach emphasizes maintaining semantic consistency across modalities. To address the complexity of label relationships, GNN4CMR^[46] employs dual generative adversarial networks to project multimodal data into a common representation space. It further utilizes multi-hop graph neural networks to model label relation dependencies and develop inter-dependent classifiers, effectively capturing complex se-

semantic structures. RONO^[47] takes a more adaptive approach by distinguishing between clean and noisy samples, providing positive and negative optimization directions, respectively. It introduces a shared space consistency learning (SSCL) mechanism to capture intrinsic information within noisy data by minimizing the cross-modal and semantic discrepancies between the common space and the label space. DRCL^[48] focuses on modality-specific quality variations by learning a transformation weight matrix for each modality. It selects the best transformation based on a quality score as the prior, avoiding indiscriminate selection of priors learned from low-quality modalities and ensuring high-quality representations.

These advances collectively broaden the scope of CCMR and enhance its adaptability to diverse modalities.

4 Fine-Grained Cross-Modal Retrieval

As discussed above, CCMR selects the most relevant coarse-grained instance according to the query. In contrast, FCMR aims to retrieve a more fine-grained target. This evolution in retrieval granularity follows two main paths. The first is subcategory-level

retrieval, including subcategory-level image retrieval, which is designed specifically for images, and subcategory-level multimodal retrieval, which enables mutual retrieval across multiple modalities. The retrieval unit remains the whole instance, but the challenge lies in distinguishing between subordinate-level fine-grained categories with subtle differences. The second involves pinpointing a specific part of an instance. This includes image grounding (IG), which retrieves a specific region in a given image, and video temporal grounding (VTG), which retrieves the precise fragment boundary in a given video, both based on a sentence description.

4.1 Subcategory-Level Image Retrieval

4.1.1 Non-VLP-Based Methods

Non-VLP-based methods for subcategory-level fine-grained image retrieval typically learn a purely visual embedding with category supervision on unimodal backbones (e.g., ResNet-50), and perform retrieval by nearest-neighbor matching in the learned metric space. As shown in Fig.5(a), these approaches can be broadly grouped into localization-based meth-

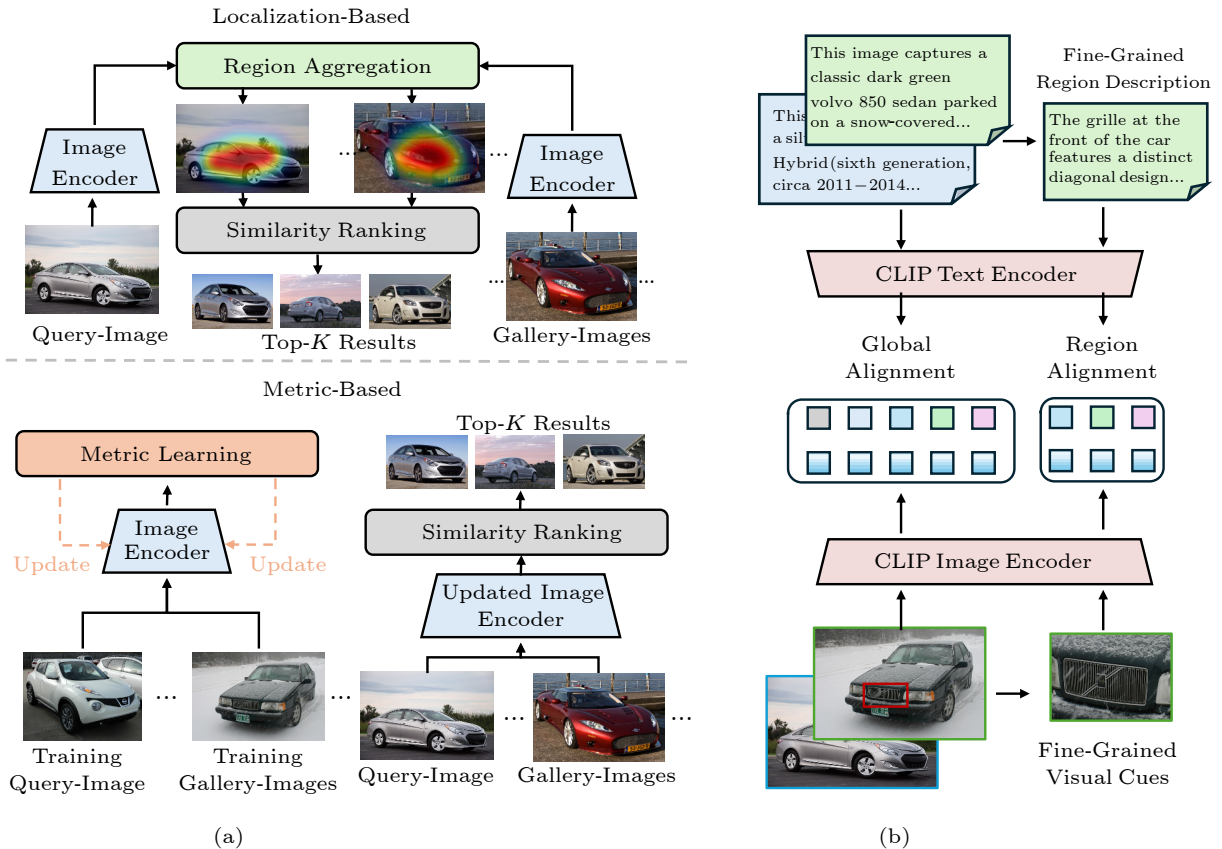


Fig.5. Illustration of typical subcategory-level image retrieval methods. (a) Non-VLP-based. (b) VLP-based.

ods that emphasize discriminative parts before pooling, and metric-based methods that optimize embedding geometry via ranking/proxy objectives. Since they mainly rely on predefined category supervision, the learned representations may entangle category-specific semantics, which can hinder generalization when the target label space changes or contains unseen categories.

Localization-Based Methods. Localization-based methods aim to suppress background clutter by explicitly highlighting the object or its discriminative regions before forming the retrieval embedding. CRL^[49] further integrates weakly supervised saliency/contour cues into CNN response maps to extract features more precisely within the target object, while pairing this with an efficient centralized ranking objective. FRPT^[50] steers a frozen pretrained model toward fine-grained retrieval by introducing a discriminative perturbation prompt that amplifies subtle differences and a lightweight category-aware adaptation head that reduces inter-class discrepancies in the feature space. For improving generalization beyond seen subcategories, CaRA^[51] rectifies overly-salient activations by suppressing dominant regions while spreading activation to adjacent non-salient areas via a content-aware prototype, encouraging more diverse discriminative cues. AdvRF^[52] formulates FGIR as an adversarial discrepancy reconstruction feedback loop, in which discrepancy localization from the retrieval model and category-agnostic feature learning from a reconstruction model mutually reinforce each other and are distilled into a robust retrieval representation.

Metric-Based Methods. Metric-based methods primarily improve fine-grained retrieval by explicitly shaping the embedding geometry through better objectives, proxy regularization, and training strategies, rather than relying on explicit region localization. HIST^[53] formulates deep metric learning as a hypergraph problem and proposes a semantic tuple loss that leverages multilateral sample-to-class relations within a mini-batch, going beyond pairwise constraints. CBML^[54] derives a Bayesian contrastive loss by modeling posterior label probabilities conditioned on feature similarity, and further adds a metric-variance constraint to improve generalization to unseen classes. NIR^[55] regularizes proxy-based learning by enforcing non-isotropic sample distributions around proxies via normalizing-flow modeling, which helps preserve local structure that is otherwise lost when optimizing only sample-proxy distances. DAS^[56] tar-

gets the “missing embedding” issue in sampling-based DML by densely synthesizing informative embeddings around anchors (via single-anchor scaling and multi-anchor shifting), thereby improving the quality of mined training pairs. HSE^[57] augments supervision by generating hybrid species through mixed-sample data augmentations, providing additional training signals when labeled samples are limited. IDML^[58] explicitly models ambiguity by representing each image with both a semantic embedding and an uncertainty embedding, and defines an introspective similarity metric that accounts for uncertainty during pairwise comparisons.

4.1.2 VLP-Based Methods

In fine-grained image retrieval, CLIP-style vision-language pre-training (VLP) models have emerged as strong zero-shot baselines. However, the subcategory-level fine-grained retrieval relies on local and subtle visual cues under large intra-class variation, where vanilla VLP models often struggle: their global image-text alignment can be disproportionately driven by background clutter, pose, or the most salient regions rather than the truly discriminative parts. To bridge this gap, recent advances have shifted focus from global alignment to fine-grained region alignment.

To explicitly enhance visual discriminability, FineCLIP^[59] moves beyond coarse-grained global alignment by leveraging region-based contrastive learning and self-distillation to encourage locally informative features, which improves fine-grained representation and makes the visual backbone more sensitive to subtle part-level variations. Following a similar region alignment manner, FLAIR^[60] learns caption-specific image representations by using the caption as a query to attend over local image tokens, which is called text-conditioned attention pooling. It also formalizes an evaluation setting where short captions describe local/partial content, explicitly testing whether the model can match fine-grained, localized semantics. On the language side, FineLIP^[61] attributes CLIP’s fine-grained bottleneck to limited text modeling capacity. It extends CLIP toward longer-text understanding and promotes more precise token-level cross-modal alignment, enabling retrieval to better reflect fine-grained textual semantics (e.g., attributes and part descriptions) rather than collapsing to broad category-level matching.

Beyond architectural modifications for region

alignment, some methods leverage prompting and reference-based refinement. From a complementary text-to-image supervision perspective, MMRef^[62] targets a practical obstacle in fine-grained retrieval: captions can be ambiguous or omit the most discriminative details. It constructs per-object multimodal references that aggregate visual and textual evidence to guide representation learning, and further applies reference-based refinement during inference to improve ranking quality, mitigating cases where the raw text query underspecifies the fine-grained distinction.

4.2 Subcategory-Level Multimodal Retrieval

While the aforementioned subcategory-level fine-grained retrieval methods primarily focus on image-text retrieval, recent advancements have expanded the scope to retrieving specific fine-grained subcategories from multimodal contents, including images, text, videos, and audio. These methods enable generalized cross-modal retrieval among any combination of these modalities.

FGCrossNet^[63] formulates fine-grained cross-media retrieval as a unified multimodal deep representation learning problem. It jointly learns representations under classification, center, and ranking constraints to extract highly discriminative and subtle common embeddings that are essential for identifying specific subcategories across images, text, videos, and audio. MGAH^[64] introduces an unsupervised multi-pathway generative adversarial hashing framework. By modeling correlation graphs, it exploits the underlying cross-modal manifold structure to capture the delicate neighborhood relationships necessary for preserving subcategory-level semantic similarities in a shared Hamming space. CDCR^[65] proposes a cross-media deep compression and regularization framework. Through media-adaptive network pruning and partial weight sharing, it enables efficient parameter learning while maintaining the precise discriminative power required for subcategory-level alignment in large-scale multimedia datasets. FGCMR^[66] introduces a fine-grained framework featuring dedicated networks for extracting modality-specific features and a common network for learning shared features across four media types. This structure jointly produces precise, discriminative representations for fine-grained categories. SAFGCMHN^[67] proposes a fine-grained cross-media retrieval framework using local self-attention layers. By combining similarity concatenation

and local position encoding, it focuses on capturing the localized semantic and spatial relationships that are crucial for identifying subcategory-specific traits, thereby reducing the feature distribution gaps across modalities. DBFC-Net^[68] introduces a double-branch network that incorporates media-specific information to construct shared features. Coupled with an improved cosine-based metric, it better handles fine-grained similarities, resulting in more accurate cross-media representations. CMCM^[69] addresses fine-grained cross-modal retrieval by performing channel mixing across deep activations. This process, enhanced by a novel fine-grained cross-modal center loss, forces the model to concentrate on the most informative feature channels for subcategory discrimination, ensuring high intra-class compactness and inter-class separability. LAGC^[70] addresses fine-grained retrieval by employing a local and global cross self-attention (LAGC-Attention) module. This allows the model to fuse subtle neighboring cues with global context, producing robust shared representations that highlight the minute differences between similar subcategories. Based on the LAGC method, LAGC-MMC^[71] further employs a unified encoder architecture and contrastive learning to align fused features at the subcategory level, ensuring that even with four or more modalities, the model maintains high sensitivity to fine-grained semantic details.

In conclusion, while both coarse-grained retrieval and subcategory-level fine-grained retrieval operate on entire instances, their objectives are fundamentally different. Coarse-grained retrieval succeeds by matching broad, high-level semantics. In contrast, subcategory-level fine-grained retrieval focuses on subtle, discriminative attributes that distinguish highly similar categories. This distinction places a greater demand on the model's ability to identify and align very specific local features and relational details, making the problem of hard-negative discrimination significantly more acute than in coarse-grained scenarios.

4.3 Image Grounding

Image grounding (IG) is a key task in fine-grained instance-part retrieval that focuses on the localization and retrieval of image regions. It takes an image and a sentence query as input to retrieve and locate the relevant object-level instance in the image. This form of retrieval requires the system not only to understand the query's semantics but also to perform a

fine-grained spatial search within the image to retrieve the corresponding region. Regrettably, acquiring manual region-sentence annotations is a costly process. Therefore, the weakly-supervised image grounding setting was proposed, where only image-sentence annotations are required. In this section, recent studies on fully-supervised IG and weakly-supervised IG are discussed.

4.3.1 Fully-Supervised Image Grounding

As shown in Fig.6, previous work can mainly be categorized into three paradigms: non-VLP-based methods, VLP-based methods, and MLLM-based methods.

1) Non-VLP-Based Methods

Non-VLP-based methods typically rely on task-specific architectures trained from scratch or fine-tuned from unimodal backbones. They can be further divided into two-stage methods, one-stage methods, and transformer-based methods.

Two-Stage Methods. Two-stage methods effectively leverage CCMR progress by transforming fine-grained localization into a proposal-based coarse-grained retrieval problem. DDPN^[3] proposes to gener-

ate proposals with more diversity and discrimination. CMRE^[72] adaptively highlights objects and the relationships between objects, such as spatial and semantic relations, which are matched with the entities in referring expressions using cross-modal attention. Two-stage methods make the pipeline clear and explainable. However, the overall performance is fundamentally capped by the quality of the initial proposals. Besides, the reliance on external detectors can also make them slow, computationally expensive due to the need to process numerous proposals, and less adaptable to domains not covered by the detector's training data.

One-Stage Methods. One-stage methods are proposed to address the limitations of two-stage methods discussed above by directly predicting the location that best matches the sentence query, without the need for a proposal generation stage. MCN^[73] proposes a multi-task framework to jointly learn referring expression comprehension (REC) and segmentation (RES), mitigating the prediction conflict between them.

Transformer-Based Methods. Recently, a series of transformer-based approaches have demonstrated the strong capability of transformers in solving the IG

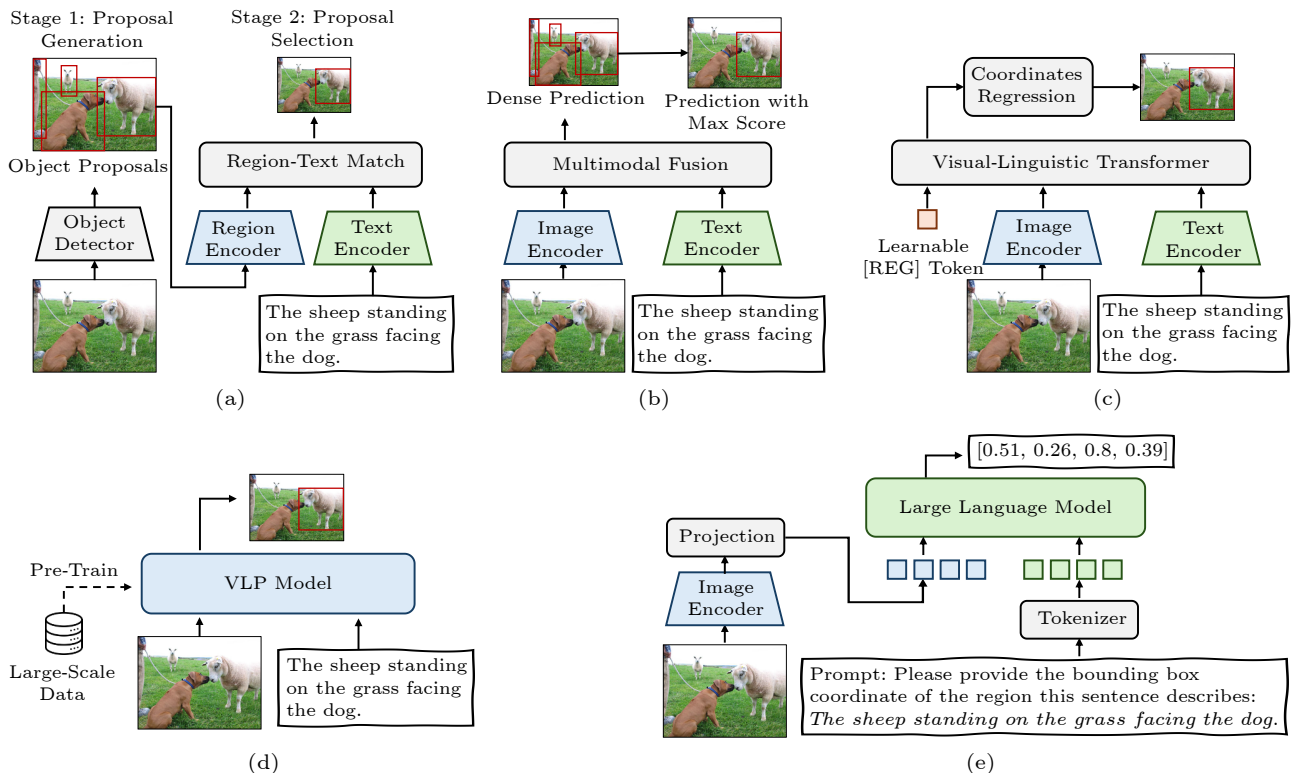


Fig.6. Illustration of five typical image grounding methods. (a) Two-stage non-VLP-based. (b) One-stage non-VLP-based. (c) Transformer-based non-VLP-based. (d) VLP-based. (e) MLLM-based.

task^[76]. While many transformer-based approaches are architecturally one-stage, we discuss them as a distinct category due to their significant methodological shift. TransVG++^[74] further employs a pure transformer architecture and leverages expressions to obtain more useful visual features, building upon TransVG. MMCA^[75] enables a visual encoder to become text-aware by dynamically constructing a low-rank weight update for its parameters, which is conditioned on a gated fusion of both visual and textual features. ResVG^[76] improves robustness to multiple-instance distractions by incorporating semantic priors from text-to-image generation and relation-sensitive data augmentation for better semantic and spatial understanding. One-stage and transformer-based methods offer significant improvements in speed and efficiency by formulating visual grounding as a direct regression or dense prediction problem, eliminating the need for a separate proposal generation step.

2) VLP-Based Methods

Lately, several VLP-based methods have been proposed for IG, achieving impressive performance^[77, 78]. These methods are pre-trained on multiple large-scale datasets and then fine-tuned on the IG task. To bridge the task gap between pre-training and visual grounding, HiVG^[77] progressively fine-tunes a model's features from shallow to deep layers using a novel Hierarchical Low-Rank Adaptation (HiLoRA) paradigm. OneRef^[78] introduces a minimalist one-tower framework that unifies expression grounding and segmentation by proposing Mask Referring Modeling (MRefM), a novel pre-training paradigm to explicitly model fine-grained referential relationships within a shared feature space. C³VG^[79] tackles ambiguities between localization and segmentation by enforcing cross-task consistency through a two-stage coarse-to-fine framework with consistency constraints and multimodal fusion. By leveraging extensive pre-training on large-scale vision-language datasets, these methods achieve top performance. Their primary strength lies in learning a robust shared embedding space, which provides strong zero-shot generalization for grounding novel objects and expressions. However, these models typically demand significant computational resources for both training and inference. Furthermore, their effectiveness is fundamentally tied to the quality of their pre-trained backbones and the sophistication of their cross-modal fusion architecture.

3) MLLM-Based Methods

With the advent of OpenAI's ChatGPT, LLMs have showcased impressive AI capabilities enabled by large-scale pre-training. Following the release of GPT-4^[110], general multimodal AI became feasible. As a result, since 2023, extensive efforts have been dedicated to grounding multimodal large language models (GM-LLMs)^[80–84].

Chen *et al.*^[82] proposed Shikra, which was pioneering in exploring GMLLMs, and conducted experimental verification, demonstrating its superior performance by directly using numerical coordinates as textual vocabulary. Grounding-GPT^[83] builds upon the Q-former^[28] frameworks to enable not only visual grounding, but also video and audio grounding. LION^[81] enhances MLLMs by injecting dual-level visual knowledge through a progressive, multi-stage training strategy for fine-grained grounding and a novel soft-prompting method for high-level semantic tags. Ferret-v2^[80] introduces a multi-granularity encoding strategy using both global context and high-resolution local details, coupled with a novel three-stage training paradigm for improved referring and grounding. CogVLM^[84] enables deep vision-language fusion without catastrophic forgetting by inserting a trainable visual expert module into each layer of a frozen pre-trained language model, achieving state-of-the-art performance across diverse multimodal tasks. Migician^[85] introduces an end-to-end framework for multi-image grounding, supported by a large-scale dataset (MGrounding-630k) and a dedicated benchmark (MIG-Bench).

MLLM-based methods represent a paradigm shift, offering unique advantages over traditional approaches. 1) *Open-Vocabulary and Commonsense Understanding*. MLLMs can interpret novel concepts and generalize to unseen entities (zero-shot capabilities), which is vital for real-world grounding tasks involving diverse and often rare objects. 2) *Instruction Following and Advanced Reasoning*. Unlike previous models, MLLMs can understand and execute complex, multi-step instructions embedded in natural language queries. This allows them to handle nuanced grounding queries that involve multiple objects, attributes, and spatial relationships. 3) *Dialogue and Iterative Refinement*. Some MLLM-based approaches can engage in dialogue to clarify ambiguous queries or iteratively refine grounding results, mimicking human-like interaction. This makes them highly adaptable to user-centric retrieval scenarios.

However, MLLMs typically employ computationally intensive “early fusion” architectures, leading to

an increased computational cost. This also makes them mainly used in FCMR, where the search space is limited to a single instance, and less prevalent in CCMR due to the prohibitive cost of applying such a heavy model across massive candidates.

Summary. The development of image grounding methods reflects an evolution from multi-step to end-to-end pipelines. Two-stage methods offer a clear pipeline but are limited by proposal quality and are less efficient. One-stage and transformer-based methods improve efficiency through direct prediction, with the latter significantly boosting performance. VLP-based and MLLM-based methods leverage large-scale pre-trained knowledge to achieve state-of-the-art performance and generalization. MLLMs, in particular, can understand complex natural language instructions, but both approaches introduce substantial computational overhead.

4.3.2 Weakly-Supervised Image Grounding

In the weakly-supervised setting, only image-text annotations are available, without paired annotations between object instances and their corresponding text descriptions. This paradigm involves training models using only coarse-grained annotations (image-text pairs), while aiming to achieve fine-grained retrieval capabilities (e.g., localizing specific regions) during inference. This setup greatly reduces annotation costs, as annotating object bounding boxes is typically time-consuming.

PSRN^[87] introduces a progressive, two-level semantic reconstruction process that first matches core “key triads” and then richer “adaptive phrases”, using a novel bidirectional supervision mechanism between the two stages. Additionally, CPL^[86] leverages pre-trained models and heuristic rules to generate text

descriptions for detected regions, mitigating the textual-visual gap. PPT^[88] introduces learnable prompts trained via novel homo-part and hetero-part contrastive losses to guide the model’s focus to different object parts, which are then fused for comprehensive localization.

Compared with fully-supervised approaches, weakly-supervised image grounding methods can establish region-query correspondence using only image-text pairs for training, thus reducing the reliance on labor-intensive bounding box annotations required in fully-supervised settings.

4.4 Video Temporal Grounding

Video temporal grounding, also known as video moment retrieval^[8, 9], refers to the task of retrieving a specific segment within an untrimmed video based on a given sentence query. This task pushes fine-grained retrieval into the temporal dimension, demanding the system to parse the video’s content over time and retrieve the precise start and end points that correspond to the textual description. To alleviate the labor-intensive process of annotating ground truth segment boundaries, weakly-supervised settings have emerged, where only video-text annotations are available. In the following, we discuss both fully-supervised and weakly-supervised VTG, similar to [Subsection 4.3](#).

4.4.1 Fully-Supervised Video Temporal Grounding

As shown in [Fig.7](#), previous work could be mainly grouped into non-VLP-based methods and MLLM-based methods.

1) Non-VLP-Based Methods

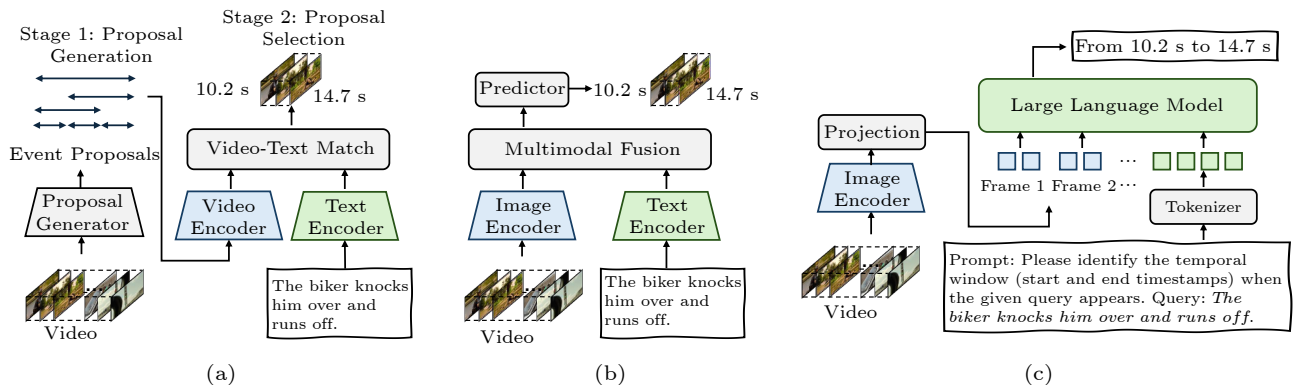


Fig.7. Illustration of three typical video temporal grounding methods. (a) Two-stage non-VLP-based. (b) One-stage non-VLP-based. (c) VLP-based.

Non-VLP-based methods typically rely on task-specific models that learn from scratch or use fixed pre-extracted features. They can be further categorized into two-stage methods and one-stage methods.

Two-Stage Methods. Two-stage methods involve a separate proposal generation stage and a proposal selection stage. To reduce the number of proposals, QSPN^[90] proposes temporal segments conditioned on the query. SAP^[89] generates a proposal according to the visual-semantic correlation score and threshold. The two-stage pipeline is conceptually straightforward, separating the task into proposal generation and subsequent ranking. However, it is inefficient due to redundant computation of largely overlapping candidate segments. More importantly, the overall performance is fundamentally limited by the quality of the initial proposals; if the correct temporal segment is not generated in the first stage, the model will inevitably fail.

One-Stage Methods. One-stage methods involve only one forward pass, without handling heavily overlapping candidate segments that are generated before. UVCOM^[92] jointly optimizes moment retrieval (MR) and highlight detection (HD) with a unified framework that integrates multi-granularity features, employing BMRW for local relation modeling in MR and a transformer encoder for global knowledge in HD. UniVTG^[91] proposes a unified framework that converts different label types into a single format, enabling large-scale pre-training and strong zero-shot generalization. One-stage methods are significantly more efficient as they eliminate the separate proposal generation step and predict boundaries in a single forward pass. This one-stage approach avoids the proposal bottleneck and allows for a more holistic understanding of the video context. OnVTG^[111] proposes a hierarchical event memory to improve the accuracy and latency for online video temporal grounding.

2) MLLM-Based Methods

The recent advancement in MLLMs has significantly enhanced the field of VTG. With their superior multimodal comprehension and reasoning abilities, MLLM-based VTG approaches are progressively surpassing traditional methods. They not only achieve competitive performance but also excel in generalization across zero-shot, multi-task, and multi-domain settings. These methods can be categorized into two main paradigms: those that require large-scale pre-training and those that operate in a training-free manner.

Pre-training approaches aim to equip models with robust temporal reasoning capabilities through large-

scale supervised learning, often involving multi-stage, progressive training pipelines before their applications to downstream tasks. TimeChat^[93] creates the TimeIT dataset with rich timestamp annotations and utilizes a Q-Former conditioned on temporal descriptors for time-sensitive long video understanding. VTimeLLM^[94] employs a boundary-aware three-stage training process that progressively refines the model’s capabilities from feature alignment to precise boundary optimization. TRACE^[96] compresses dense visual inputs using slot embeddings and adds task-specific decoding heads to produce structured temporal outputs through causal event modeling. TimeSuite^[95] provides a unified collection of diverse datasets to facilitate comprehensive temporal learning and improve long video understanding via grounded tuning. LLaVA-ST^[97] employs a hierarchical framework and incorporates temporal position embeddings to advance fine-grained spatial-temporal understanding. TimeR1^[98] introduces a reasoning-guided framework that adapts reinforcement learning based strategies with a novel reward mechanism to enhance temporal reasoning.

Training-free approaches leverage pre-trained foundation models and specialized expert tools, enabling temporal grounding through purely inference-based pipelines without requiring labeled supervision or end-to-end training. ChatVTG^[112] employs MLLM to generate multi-granularity clip captions and then facilitates iterative query matching. VTG-GPT^[113] is a framework that leverages GPT for temporal grounding by matching generated video descriptions to textual queries. TFVTG^[114] proposes a training-free zero-shot pipeline for VTG by decomposing complex queries into sub-events using an LLM and then performing segment matching using a VLM.

MLLM-based approaches offer distinct advantages. 1) *Open-Vocabulary and Commonsense Understanding.* MLLMs can generalize to unseen events and actions, vital for real-world videos. 2) *Instruction Following and Advanced Reasoning.* MLLMs can execute complex, multi-step instructions, such as “Find the moment where the person in red is running, but before they fall”. 3) *Dialogue and Iterative Refinement.* Systems like ChatVTG can interact with users to refine the temporal boundaries.

While MLLM-based approaches excel in zero-shot performance and generalization, their high computational cost hinders practical application, especially in resource-constrained environments. Furthermore, MLLM context length and efficiency limitations pose challenges for processing long videos.

Summary. The evolution of fully supervised video temporal grounding methods also shows a progression in efficiency and performance. Two-stage methods break the task into proposal generation and selection; this approach is conceptually simple but inefficient, and its performance is capped by the quality of the initial proposals. One-stage methods, through end-to-end prediction, are not only more efficient but also demonstrate clear performance advantages over two-stage approaches. Meanwhile, MLLM-based methods show significant advantages in both fully-supervised and zero-shot scenarios due to their powerful reasoning capabilities, but their high computational cost and limitations in processing long videos are major current challenges.

4.4.2 Weakly-Supervised Video Temporal Grounding

The large-scale annotations of VTG are labor-intensive and costly. Besides, the boundary of temporal annotation might be inaccurate due to the continuous transition of the video. Therefore, weakly-supervised VTG is proposed to mitigate the issue above, where only a video-text pair is provided.

CNM^[99] employs a learnable Gaussian mask to generate proposals and an intra-video contrastive loss to distinguish the semantics of the segment better. Similar to CNM^[99], CPL^[100] utilizes learnable Gaussian functions to generate proposals, where the hard negative proposals are used to distinguish confusing segments from the target segment. SCANet^[101] measures the complexity of video scenes to obtain a video-adaptive proposal, considering that the number of scenes varies for different videos. PPS^[102] introduces expressive Gaussian mixture proposals, trained with a novel pull-push scheme, to capture diverse, multi-part events for weakly-supervised temporal video grounding.

Weakly-supervised video temporal grounding reduces annotation costs by avoiding the need for precise start and end timestamp labels, which are notoriously expensive and subjective to create. However, the lack of explicit temporal supervision presents significant challenges. Multi-instance learning (MIL) approaches often struggle because a positive video may contain many irrelevant frames (false positives), making it difficult to isolate the true event segment. Reconstruction-based methods can also fail, as powerful models may successfully reconstruct a query using features from an incorrect video segment.

4.5 FCMR for Other Modalities

Fine-grained cross-modal retrieval is not limited to image-text or video-text scenarios; it can also be extended to a wide range of other modalities, such as audio and 3D data. These extensions unlock new possibilities and broaden the applicability of FCMR in diverse domains. For example, 3D visual grounding^[106–109] has emerged as an important task in cross-modal understanding, aiming to associate language queries with specific regions in 3D scenes. Recent advances have introduced dedicated 3D feature extractors and encoders to better capture geometric and semantic relationships within 3D data. These developments have improved the precision and robustness of 3D visual grounding, enabling more comprehensive scene understanding in real-world applications.

5 Datasets and Evaluation

The detailed information of all datasets used for cross-modal retrieval in this paper is summarized in Table 2, with specific details also available in the respective subsection of each section. It should be noted that due to different task settings (like grounding), the data volume of the same dataset may vary across different tasks, such as Flickr30k^[116].

5.1 Image-Text Retrieval

5.1.1 Datasets

In the field of image-text retrieval (ITR), model performance is commonly evaluated on two frequently used and publicly available datasets: Flickr30k and the Microsoft Common Objects in COntext (MS COCO) dataset. Each image in these two datasets is described with about five different corresponding sentence descriptions.

Flickr30k^[116]. It collects 31 783 images in total, and is divided into three parts: the training set with 29 783 images, the validation set with 1 000 images, and the test set with 1 000 images.

MS COCO^[115]. This dataset is more large-scale than Flickr30k, and contains 123 287 images in total.

In addition to these classic benchmark datasets for evaluation, recent years have seen the creation of several large-scale, Internet-sourced datasets to support the pre-training. These datasets are characterized by their massive scale, though they often contain noise.

Table 2. Detailed Statistics for Cross-Modal Retrieval Datasets

Task	Dataset	Number of Instances (k)	Number of Queries (k)	Object Category	Modality	Main Scene
Image-text retrieval	MS COCO ^[115]	123.300	616.4	80	Text, image	Scenery, human
	Flickr30k ^[116]	31.800	158.9	–	Text, image	Scenery
Video-text retrieval	MSR-VTT ^[117]	10.000	200.0	20	Text, video	Human
	ActivityNet ^[118]	20.000	20.0	–	Text, video	Object, human
	LSMDC ^[119]	118.100	118.1	–	Text, video	Movie
	Vatex ^[120]	41.300	825.0	600	Text, video	Human
	PKU XMedia ^[121]	12.000	—	20	Text, image, video, audio, 3D	Animal
Multimodal retrieval	PKU XMediaNet ^[122]	100.000	—	200	Text, image, video, audio, 3D	Animal, artifact
Image grounding	RefCOCO ^[123]	50.000	142.2	–	Text, image	Scenery, human
	RefCOCO+ ^[123]	49.900	141.6	–	Text, image	Scenery, human
	RefCOCOg ^[124]	49.800	95.0	–	Text, image	Object, human
	ReferItGame ^[125]	20.000	120.1	–	Text, image	Scenery, human
	Flickr30k Entities ^[116]	427.000	456.1	–	Text, image	Scenery
Video temporal grounding	TACoS ^[126]	0.126	18.2	–	Text, video	Human
	MAD ^[127]	0.654	384.6	–	Text, video	Movie
	ActivityNet Captions ^[118, 128]	15.000	72.0	–	Text, video	Object, human
	Charades-STA ^[129]	6.600	16.1	–	Text, video	Indoor, human
Subcategory-level fine-grained retrieval	PKU FG-XMedia ^[63]	50.138	—	200	Text, image, video, audio	Bird
	CUB-200-2011 ^[130]	11.788	—	200	Text, image	Bird
	Stanford Cars ^[131]	16.185	—	196	Text, image	Car
	FGVC-Aircraft ^[132]	1.020	—	102	Text, image	Aircraft

CC12M^[133]. It contains approximately 12 million image-text pairs that are automatically collected from the web. It is primarily used as a large-scale pre-training resource to enhance cross-modal representation learning.

LAION-5B^[134]. It contains approximately 5.85 billion image-text pairs collected from the Common Crawl^① and filtered by the CLIP model, making it one of the largest open-access image-text datasets. Its immense scale and diversity have been pivotal for training powerful foundation models.

DataComp^[135]. It serves not only as a dataset but also as a platform for researching dataset construction methodologies. Researchers created multiple versions of DataComp (e.g., DataComp-1B) from a massive pool of 12.8 billion image-text pairs by applying different filtering strategies.

5.1.2 Evaluation Metrics

Recall@K ($R@K$). This is a widely used metric for evaluating the performance of CCMR. It measures the

percentage of ground truth instances correctly retrieved within the top- K results.

Mean Average Precision (mAP). mAP is widely used in hash-based image-text retrieval. mAP evaluates the average precision over all queries. While Recall@ K focuses on the hit rate (whether the target appears in the top K results), it does not account for the ranking position of the correct results within that list or the precision density. mAP penalizes relevant items that are ranked lower in the list and thus provides a more comprehensive measure of the overall ranking quality.

5.1.3 Performance Comparison

The results of feature alignment methods and VLP-based methods are shown in Table 3, while the performances of hash encoding methods are listed in Table 4. Global alignment based methods are inferior to the local alignment based methods; even the best global alignment based method ESSE^[19] still lags behind the local alignment based methods. It indicates

^①<https://commoncrawl.org/>, Jan. 2026.

Table 3. Performance Comparison of Image-Text Retrieval Methods on MS COCO (1k), MS COCO (5k), and Flickr30k Datasets

Model	Source	MS COCO (1k)						MS COCO (5k)						Flickr30k					
		Image-to-Text			Text-to-Image			Image-to-Text			Text-to-Image			Image-to-Text			Text-to-Image		
		R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
ESSE ^[19]	TMM 2024	69.4	92.5	—	56.1	94.6	—	45.9	75.2	—	33.4	63.3	—	—	—	—	—	—	—
3SHNet ^[23]	IPM 2024	85.8	97.7	99.3	71.8	93.7	97.4	67.1	89.8	95.2	49.9	78.8	87.2	86.1	97.6	98.8	68.6	90.1	94.4
TVRN ^[24]	TMM 2024	81.1	96.4	98.8	67.7	92.3	97.1	61.1	86.3	92.5	45.5	75.0	84.8	79.7	95.4	97.6	62.3	85.3	90.6
GFRAN ^[25]	PR 2025	83.0	96.4	98.9	68.7	94.6	97.6	65.9	86.1	92.7	46.3	75.8	84.8	85.6	96.3	98.7	71.3	91.3	95.6
CLIP ^[26]	ICML 2021	—	—	—	—	—	—	58.4	81.5	88.1	37.8	62.4	72.2	88.0	98.7	99.7	75.7	93.8	96.8
BEIT-3 ^[27]	CVPR 2023	—	—	—	—	—	—	84.8	96.5	98.3	67.2	87.7	92.8	94.9	99.9	100.0	81.5	95.6	97.8
BLIP-2 ^[28]	ICML 2023	—	—	—	—	—	—	85.4	97.0	98.5	68.3	87.7	92.6	97.6	100.0	100.0	89.7	98.1	98.9
LAPS ^[29]	CVPR 2024	—	—	—	—	—	—	72.9	91.7	—	57.1	81.3	—	94.6	99.9	—	84.9	97.3	—
USER ^[30]	TIP 2024	83.7	96.7	99.0	67.8	91.2	95.8	67.6	88.4	93.5	47.7	75.1	83.7	86.3	97.6	99.4	69.5	91.0	94.4
MV-VSE++ ^[31]	PR 2025	83.8	97.0	98.9	68.8	91.6	96.5	68.5	88.9	93.5	50.2	75.3	83.8	92.7	99.5	99.7	78.8	95.3	97.6

Note: MS COCO (1k) and MS COCO (5k) denote the subsets of 1 000 and 5 000 images, respectively, from the MS COCO test set. Higher metric values indicate better. The first four models are non-VLP-based, and the others are VLP-based.

Table 4. Performance Comparison of Methods Using Hash Encoding in the Image-Text Retrieval Task with Multiple Lengths of Hash Codes

Model	Source	Image-to-Text (mAP)				Text-to-Image (mAP)			
		16 Bits ↑	32 Bits ↑	64 Bits ↑	Mean ↑	16 Bits ↑	32 Bits ↑	64 Bits ↑	Mean ↑
DSPH ^[22]	TCSVT 2023	70.44	75.10	76.94	74.16	70.40	75.56	77.13	74.36
DNPH ^[21]	TOMM 2024	64.38	69.10	72.94	68.81	64.68	70.12	73.88	69.56
DIMCH ^[20]	TIP 2025	73.98	77.26	78.45	76.56	74.82	77.88	79.84	77.51

Note: ↑ means higher metric values indicate better.

the significance of fine-grained information in the image and text, which is a supplement to coarse-grained knowledge. Meanwhile, we have witnessed the continuous improvement of local alignment based methods. Besides, VLP-based methods have become dominant in the task, which are better than typical methods. The success of VLP-based methods mainly comes from the large-scale data, that is to say, they are data-driven. The recently proposed model DIMCH^[20] using hash encoding achieves significant improvements over other methods, which shows the effectiveness of multi-granularity alignments in ITR.

5.2 Video-Text Retrieval

5.2.1 Datasets

The performance evaluation for video-text retrieval is conducted on several standard datasets.

MSR-VTT^[117]. It contains 10 000 videos and corresponding annotated 200 000 sentences. 1K split is the most common split, where 9 000 videos are for training, and the remaining 1 000 videos are used to validate the performance of the proposed method.

ActivityNet Captions^[118, 128]. It consists of 19 994

YouTube videos. The most commonly used split includes 10 009 videos for the training set and 4 917 videos (with 17 505 captions) for the evaluation set (val_1) to facilitate performance comparison.

LSMDC^[119]. It consists of 118 081 video clips from 202 movies with corresponding caption descriptions. LSMDC is usually divided into a training set with 109 673 videos, a validation set with 7 408 videos, and a test set with 1 000 videos.

Vatex^[120]. It is a multilingual video description dataset containing over 41 250 videos and 825 000 captions. Typically, 25 991 videos are used for training and 1 500 videos for testing.

Similar to image-text retrieval, the video-text retrieval field has benefited from several large-scale web-sourced datasets, which are key for training modern video-language models.

InternVid^[136]. It is a large-scale video-text dataset built for training video foundation models, containing over 230 million video clips and 7 billion text descriptions.

WebVid^[137]. It is an earlier large-scale dataset that contains approximately 10 million video-caption pairs scraped from the web, featuring a wide diversity of short-form content.

5.2.2 Evaluation Metrics

In addition to Recall@ K introduced in image-text retrieval, median rank (MdR) and mean rank (MnR) are another two commonly used metrics.

MdR is the median rank of the first correct result across all queries, while MnR is the arithmetic mean of the ranks (lower is better). The rank is the position of the correct result within the retrieved items.

5.2.3 Performance Comparison

The video-text retrieval performance comparisons on MSR-VTT and the other three datasets are shown in Table 5 and Table 6, respectively, where $\uparrow$ means higher metric values indicate better and $\downarrow$ is the opposite. Recent VLP-based methods show their superiority over non-VLP-based methods. Notably, LEAN^[33] has a comparable performance with some VLP-based methods. There is a continuous trend of performance increment for VLP-based methods in recent few years. In the VLP-based methods, CLIP-based methods have obtained the dominant attention with the benefit of efficiency. With knowledge transfer and adapter, CLIP-based methods like MMI-TVR^[36] and JM-CLIP^[37] achieve high performances with relatively low

computational cost, based on the large-scale information in CLIP pre-trained using 400 million data.

5.3 Multimodal Retrieval

5.3.1 Datasets

Datasets for multimodal retrieval tasks include PKU XMedia^[121] and PKU XMediaNet^[122], which are specifically designed for cross-media retrieval. These datasets provide diverse media types and are widely used to evaluate the performance of multimodal retrieval systems.

PKU XMedia^[121]. It is the first cross-media dataset with five media types, containing 12 000 media instances. It consists of 5 000 text descriptions, 5 000 images, 500 videos, 1 000 audio clips, and 500 3D models, all of which are collected from well-known websites. Each media instance is labeled with one of 20 categories, with 600 instances per category.

PKU XMediaNet^[122]. It is a large-scale multimodal dataset designed for cross-media retrieval tasks. It consists of 40 000 images, 40 000 text descriptions, 2 000 3D models, 10 000 audio clips, and 10 000 videos. The dataset is constructed by selecting 200 category nodes from WordNet to ensure a semantic

Table 5. Performance Comparison of Video-Text Retrieval Models on MSR-VTT 1k-A Split

Category	Model	Source	Text-to-Video					Video-to-Text				
			R@1 $\uparrow$	R@5 $\uparrow$	R@10 $\uparrow$	MDR $\downarrow$	MNR $\downarrow$	R@1 $\uparrow$	R@5 $\uparrow$	R@10 $\uparrow$	MDR $\downarrow$	MNR $\downarrow$
Non-VLP-based	LEAN ^[33]	WSDM 2024	50.6	77.1	85.2	2.0	11.4	49.1	78.2	86.7	2.0	7.3
	TV-ProxyNet ^[34]	AAAI 2025	55.2	80.4	86.8	1.0	9.3	–	–	–	–	–
VLP-based	DGL ^[35]	AAAI 2024	49.7	73.1	82.3	–	15.1	47.8	74.1	83.3	–	10.6
	MMI-TVR ^[36]	ACM MM 2024	52.4	78.0	87.6	2.0	11.1	51.2	79.5	87.4	2.0	6.8
	JM-CLIP ^[37]	ICASSP 2024	62.5	85.3	93.7	1.0	2.9	58.9	88.0	96.0	1.0	2.6
	M3T-PS ^[38]	Visual Intelligence 2025	52.0	77.1	84.9	1.0	–	–	–	–	–	–
	OFIA ^[39]	CIKM2025	54.1	81.1	96.9	1.5	–	52.9	90.1	95.2	1.0	–

Note: Existing methods are divided into non-VLP-based methods and VLP-based methods.

Table 6. Performance Comparison of Video-Text Retrieval Models on ActivityNet Caption Dataset, LSMDC Dataset, and VateX Dataset, Respectively

Dataset	Model	Source	Text-to-Video					Video-to-Text				
			R@1 $\uparrow$	R@5 $\uparrow$	R@10 $\uparrow$	MDR $\downarrow$	MNR $\downarrow$	R@1 $\uparrow$	R@5 $\uparrow$	R@10 $\uparrow$	MDR $\downarrow$	MNR $\downarrow$
LSMDC	DGL ^[35]	AAAI 2024	21.6	39.3	49.0	–	64.4	–	–	–	–	–
	MMI-TVR ^[36]	ACM MM 2024	26.6	44.6	54.1	7.0	51.8	24.3	43.2	53.9	10.0	45.3
Vatex	DGL ^[35]	AAAI 2024	57.3	87.0	93.3	–	4.2	–	–	–	–	–
ActivityNet Caption	TV-ProxyNet ^[34]	AAAI 2025	56.7	83.1	91.1	1.0	–	–	–	–	–	–
	DGL ^[35]	AAAI 2024	43.1	72.3	82.7	–	8.6	–	–	–	–	–
	MMI-TVR ^[36]	ACM MM 2024	45.3	75.1	87.9	2.0	5.7	–	–	–	–	–
	JM-CLIP ^[37]	ICASSP 2024	72.0	94.6	96.8	1.0	1.7	–	–	–	–	–

hierarchy structure.

5.3.2 Evaluation Metrics

Similar to other retrieval tasks, multimodal retrieval also relies on mAP-based metrics for performance evaluation, including mAP@50 and mAP@all.

5.3.3 Performance Comparison

The performance comparison of multimodal retrieval models on the PKU XMedia is presented in Table 7. Methods based on common subspace learning and semantic alignment generally outperform those focusing primarily on noise robustness. Specifically, the noise-robust method RONO^[47] exhibits lower mAP scores across most modality pairs. This indicates that while addressing noisy labels and samples is crucial, the fundamental ability to align diverse modalities in a shared space remains the primary driver of retrieval accuracy. Meanwhile, we observe that modeling complex dependencies provides a performance boost. The recently proposed DRCL^[48] achieves significant improvements over all the other methods, reaching the highest average mAP. The success of DRCL^[48] mainly stems from its adaptive transformation weight matrix and quality-based prior selection, which avoids the interference of low-quality modalities. This demonstrates that dynamically weighting modality contributions is a more effective strategy for large-scale multimodal retrieval than static alignment or simple noise filtering.

5.4 Subcategory-Level Fine-Grained Retrieval

5.4.1 Datasets

Datasets for subcategory-level fine-grained retrieval tasks primarily include the multimodal dataset PKU FG-XMedia^[63], which was specifically built for this purpose, as well as several classic datasets originally designed for fine-grained image classification that are now widely used to evaluate fine-grained retrieval performance, such as CUB-200-2011 and Stanford Cars.

PKU FG-XMedia^[63]. It is a large-scale, fine-grained cross-media dataset specifically designed for fine-grained retrieval. It contains 200 fine-grained categories, covering four modalities: image, video, audio,

and text. Each category includes instances of multiple media types, making it an ideal benchmark for evaluating complex cross-media retrieval tasks.

CUB-200-2011^[130]. It is a classic benchmark in fine-grained visual classification, comprising 11 788 images of 200 bird species. The training set contains 5 994 images, and the test set contains 5 794 images. Although originally intended for classification, its rich categories and subtle inter-class variations have led many studies to use it for evaluating single-modal fine-grained image retrieval performance.

Stanford Cars^[131]. It contains 16 185 images of 196 classes of cars, with categories detailed down to the make, model, and year. It is divided into a training set with 8 144 images and a test set with 8 041 images.

FGVC-Aircraft^[132]. It contains 10 200 images of aircraft, with 100 images for each of 102 different aircraft model variants, most of which are airplanes.

5.4.2 Evaluation Metrics

For subcategory-level fine-grained retrieval, the metrics used are typically the same as for coarse-grained retrieval, such as mAP and Recall@K.

5.4.3 Performance Comparison

Table 8 summarizes Recall@K performance on three standard fine-grained image retrieval benchmarks. Prompt/adaptation and generalization-driven designs such as FRPT^[50] and CaRA^[51] push performance to a higher regime, with PLEor and CaRA achieving particularly strong results on Cars and Aircraft. Finally, AdvRF^[52] establishes the best overall performance among the listed methods, delivering the strongest Recall@K across all three datasets, which suggests that explicitly emphasizing fine-grained discrepancy learning and robustness-oriented representation transfer can provide complementary benefits beyond conventional localization or metric-learning alone.

Table 9 presents the mAP performance of representative subcategory-level multimodal retrieval methods on the PKU FG-XMedia dataset. With the emergence of fine-grained multimodal representation learning, methods like CDCR^[65] significantly improve retrieval accuracy by explicitly modeling fine-grained semantic correspondences and multimodal structural relationships. DBFC-Net^[68] and CMCM^[69] improve fine-grained cross-modal retrieval through enhanced

Table 7. Performance (mAP@all) Comparison of Multimodal Retrieval Models on PKU XMedia Dataset

Method	Source	Image			Text			Audio			3D			Video			Avg.					
		Text	Audio	3D	Video	Image	Audio	Image	Text	3D	Image	Text	Audio	Image	Text	Audio						
ELRCMR ^[44]	ACM MM 2022	0.814	0.329	0.419	0.370	0.779	0.312	0.400	0.320	0.289	0.271	0.138	0.128	0.385	0.399	0.182	0.303	0.493	0.491	0.128	0.386	0.367
MARS ^[45]	TCVST 2022	0.887	0.569	0.643	0.642	0.888	0.593	0.667	0.671	0.534	0.565	0.454	0.409	0.652	0.667	<u>0.430</u>	0.488	0.584	0.605	0.381	0.473	0.590
GNN4CMR ^[46]	TPAMI 2023	0.875	0.453	0.624	0.627	0.880	0.472	0.642	0.617	0.475	0.480	0.415	0.244	0.595	0.587	0.306	0.446	0.562	0.563	0.238	0.459	0.528
RONO ^[47]	CVPR 2023	0.896	0.469	0.656	0.576	0.906	0.442	0.665	0.637	0.474	0.480	0.124	0.124	0.584	0.619	0.199	0.413	0.555	0.589	0.141	0.496	0.502
DRCL ^[48]	TMM 2025	0.902	0.624	0.698	0.662	<u>0.909</u>	0.666	0.729	0.695	0.593	0.625	0.506	0.443	0.688	0.712	0.460	<u>0.522</u>	0.594	0.621	0.421	0.539	0.630

Note: Higher metric values indicate better.

Table 8. Fine-Grained Image Retrieval Performance (%) on CUB-200-2011, Stanford Cars 196, and FGVC Aircraft with ResNet-50 Backbone

Method	Source	CUB-200-2011				Stanford Cars 196				FGVC Aircraft			
		$K = 1$	$K = 2$	$K = 4$	$K = 8$	$K = 1$	$K = 2$	$K = 4$	$K = 8$	$K = 1$	$K = 2$	$K = 4$	$K = 8$
CRL ^[49]	IJCAI 2018	62.5	74.2	82.9	89.7	57.8	69.1	78.6	86.6	61.1	71.6	80.9	88.2
FRPT ^[50]	AAAI 2023	74.3	83.7	89.8	94.3	91.1	95.1	97.3	98.6	77.6	85.7	91.4	95.6
CaRA ^[51]	TPAMI 2024	73.9	82.2	89.4	93.6	94.1	96.9	98.2	98.9	84.3	90.4	94.2	96.3
AdvRF ^[52]	ICCV 2025	76.6	85.3	91.7	95.0	94.9	97.2	98.6	98.9	88.0	92.5	95.5	96.9
HIST ^[53]	CVPR 2022	71.4	81.1	88.1	—	89.6	93.9	96.4	—	—	—	—	—
CBML ^[54]	TPAMI 2022	69.9	80.4	87.2	92.5	88.1	92.6	95.4	97.4	—	—	—	—
NIR ^[55]	CVPR 2022	70.5	80.6	—	—	89.1	93.4	—	—	—	—	—	—
DAS ^[56]	ECCV 2022	69.2	79.3	87.1	92.6	87.8	93.2	96.0	97.9	—	—	—	—
HSE ^[57]	ICCV 2023	70.6	80.1	87.1	—	89.6	93.8	96.0	—	—	—	—	—
IDML ^[58]	TPAMI 2023	70.7	80.2	—	—	90.6	94.5	—	—	—	—	—	—

Note: We report Recall@K for $K \in \{1, 2, 4, 8\}$.

Table 9. Performance (mAP) Comparison of Subcategory-Level Fine-Grained Retrieval Methods on PKU FG-XMedia Dataset

Method	Source	I→T	I→A	I→V	T→I	T→A	T→V	A→I	A→T	A→V	V→I	V→T	V→A	Average
FGCrossNet ^[63]	ACM MM 2019	0.210	0.526	0.606	0.255	0.181	0.208	0.553	0.159	0.443	0.629	0.195	0.437	0.366
MGAH ^[64]	IEEE TMM 2019	0.162	0.125	0.508	0.168	0.159	0.174	0.246	0.153	0.308	0.543	0.162	0.358	0.255
CDCR ^[65]	IEEE TCSVT 2019	0.173	0.152	0.486	0.174	0.163	0.185	0.284	0.162	0.321	0.556	0.184	0.382	0.268
FGCMR ^[66]	PRCV 2020	0.355	0.629	0.660	0.409	0.324	0.335	0.643	0.287	0.515	0.706	0.335	0.544	0.478
SAFGCMHN ^[67]	ACM MM Asia 2021	0.293	0.625	0.636	0.335	0.263	0.272	0.629	0.231	0.495	0.677	0.270	0.525	0.437
DBFC-Net ^[68]	Multimedia System 2022	0.298	0.563	0.626	0.346	0.252	0.284	0.580	0.223	0.462	0.669	0.286	0.483	0.423
CMCM ^[69]	ICME 2022	0.349	0.586	0.597	0.353	0.284	0.283	0.592	0.281	0.450	0.625	0.323	0.486	0.434
LAGC ^[70]	ICUFN 2023	0.357	0.635	0.646	0.431	0.328	0.352	0.675	0.313	0.611	0.691	0.343	0.601	0.499
LAGC-MMC ^[71]	Expert Syst Appl 2025	0.626	0.631	0.633	0.649	0.592	0.573	0.663	0.586	0.583	0.659	0.624	0.642	0.622

Note: Higher metric values indicate better. I, T, A, V mean image, text, audio, and video, respectively.

attention and similarity modeling, consistently reducing modality gaps in both one-to-one and one-to-all settings. LAGC^[70] further models local and global cross-modal dependencies via cross self-attention, while LAGC-MMC^[71] adopts a unified encoder with MMC and contrastive learning, achieving state-of-the-art results on PKU FG-XMedia. Overall, strengthening fine-grained semantics, attention-based interaction, and contrastive alignment is key to subcategory-level multimodal retrieval.

5.5 Image Grounding

5.5.1 Datasets

RefCOCO^[123]/*RefCOCO+*^[123]/*RefCOCog*^[124]. They are all derived from MS COCO^[115]. RefCOCO comprises 19 994 images containing 50 000 referred objects with 142 210 corresponding referring expressions. RefCOCO+ consists of 19 992 images with 49 856 referred objects and 141 564 corresponding referring expressions. RefCOCog contains 25 799 images with 49 822 referred objects and 95 010 corresponding referring expressions.

ReferItGame^[125]. It contains 20 000 images collected from the SAIAPR-12 dataset^[138], which is divided into a training set with 54 127 expressions, a validation set with 5 842 expressions, and a test set with 60 103 expressions.

Flickr30k Entities^[116]. As discussed in Subsection 5.1, Flickr30k Entities contains 31 783 images with 427 000 referred entities, which is divided into a training set with 29 783 images, a validation set with 1 000 images, and a test set with 1 000 images.

5.5.2 Evaluation Metrics

Intersection over Union (IoU). IoU@0.5 is employed to evaluate the performance of the IG model.

This metric calculates the percentage of queries where the top-predicted bounding box has an intersection over union (IoU) with the ground-truth box greater than a specific threshold (typically 0.5).

5.5.3 Performance Comparison

The performance of fully-supervised and weakly-supervised methods in IG is shown in Table 10 and Table 11, respectively. For the RefCOCO and RefCOCO+ dataset, the val split is the validation set, the testA split specifically evaluates the model's ability to distinguish human instances, containing images with multiple people, and the testB split focuses on evaluating the grounding of inanimate objects, containing images with multiple non-human instances. For the RefCOCog dataset, the val-g split refers to the validation set from the original Google split, which has the limitation of sharing some identical images with the training set. The val-u and test-u splits are the validation set and test set from the improved split, respectively, which guarantees a strict zero image overlap with the training set. We group fully-supervised methods into five categories: two-stage methods, one-stage methods, transformer-based methods, VLP-based methods, and MLLM-based methods. There is no significant difference in performance between two-stage methods and one-stage methods. However, one-stage methods tend to reduce the computational cost compared with two-stage methods. Transformer-based methods demonstrate superior capability in solving IG compared with the previous two approaches. VLP-based and MLLM-based methods listed in Table 10 perform better than the other methods, but the gap is less than what is shown in the ITR task (Table 3).

In a weakly-supervised setting, only image-text pairs are given without the annotations of the target location, which makes it more challenging. Notably, CPL^[86] and PPT^[88] perform well in this setting and

Table 10. Performance (IoU@0.5) Comparison of Image Grounding Models on Five Popular Datasets in the Fully-Supervised Setting

Category	Model	Source	RefCOCO			RefCOCO+			RefCOCOg			RG	FE
			val	testA	testB	val	testA	testB	val-g	val-u	test-u	Test	Test
Non-VLP-based	DDPN ^[3]	IJCAI 2018	76.80	80.10	72.40	64.80	70.50	54.10	–	–	–	63.00	73.30
	CMRE ^[72]	TPAMI 2020	–	82.53	68.58	–	75.76	57.27	–	–	67.38	–	–
	MCN ^[73]	CVPR 2020	80.08	82.29	74.98	67.16	72.86	57.31	66.46	–	66.01	–	–
	TransVG++ ^[74]	TPAMI 2023	86.28	88.37	80.97	75.39	80.45	66.28	73.86	76.18	76.30	74.70	81.49
	MMCA ^[75]	ACM MM 2024	84.76	87.34	80.86	73.18	78.67	64.13	72.53	74.91	73.87	73.43	–
	ResVG ^[76]	ACM MM 2024	85.51	88.76	79.93	73.95	79.53	64.88	73.13	75.77	74.53	72.35	79.52
VLP-based	HiVG ^[77]	ACM MM 2024	90.77	92.94	88.03	86.78	89.91	78.02	–	86.61	86.60	78.16	82.63
	OneRef ^[78]	NeurIPS 2024	92.87	94.01	90.19	87.98	91.57	83.73	–	88.11	89.29	81.11	84.75
	C ³ VG ^[79]	AAAI 2025	92.51	94.60	88.71	87.44	90.69	81.42	–	87.68	88.31	–	–
MLLM-based	Ferret-v2-13B ^[80]	arXiv 2024	92.64	94.95	88.86	87.39	92.05	81.36	–	89.43	89.99	–	86.25
	LION-14B ^[81]	CVPR 2024	89.80	93.02	85.57	83.95	89.22	78.06	–	85.52	85.74	–	85.57
	Shikra-13B ^[82]	arXiv 2023	87.83	91.11	81.81	82.89	87.79	74.41	–	82.64	83.16	–	–
	Grounding-GPT-7B ^[83]	ACL 2024	88.02	91.55	82.47	81.61	87.18	73.18	–	81.67	81.99	–	–
	CogVLM-17B ^[84]	NeurIPS 2024	92.76	94.75	88.99	88.68	92.91	83.39	–	89.75	90.79	–	–
	Migician ^[85]	ACL 2025	91.62	93.49	87.22	86.13	91.06	79.93	–	88.06	87.80	–	–

Note: RG denotes ReferitGame, and FE denotes Flickr30k Entities. Existing methods are categorized into four sets for further comparison and analysis. Higher metric values indicating better.

Table 11. Performance (IoU@0.5) Comparison of Image Grounding Models on Five Popular Datasets in the Weakly-Supervised Setting

Model	Source	RefCOCO			RefCOCO+			RefCOCOg	ReferitGame	Flickr30k Entities
		val	testA	testB	val	testA	testB	val-g	Test	Test
CPL ^[86]	ICCV 2023	70.67	74.58	67.19	51.81	58.34	46.17	57.04	45.23	63.87
PSRN ^[87]	TCSVT 2024	41.31	41.48	42.12	40.68	40.15	41.42	50.07	–	–
PPT ^[88]	MMM 2024	68.16	77.53	58.22	68.62	77.32	57.24	–	–	–

Note: Higher metric values indicate better.

surpass other weakly-supervised methods by a large margin, even approaching the performance of some fully-supervised methods. The underlying reason is that CPL adopts a pre-trained VL model BLIP^[139] and PPT introduces learnable prompts.

5.6 Video Temporal Grounding

5.6.1 Datasets

Video temporal grounding datasets mainly include ActivityNet Captions^[118, 128], Charades-STA^[129], TACoS^[126], and MAD^[127]. Among them, the two most widely used datasets are ActivityNet Captions and Charades-STA.

ActivityNet Captions^[118, 128]. The ActivityNet Captions dataset comprises 37 417, 17 505, and 17 031 video-sentence pairs in the train, val_1, and val_2 splits, respectively. The average length of the target segment and untrimmed videos are 36.2 and 117.6 seconds, respectively.

Charades-STA^[129]. The Charades-STA dataset

comprises 12 408 and 3 720 video-sentence pairs in the train and test sets, respectively. The average length of query sentences is 7.2 words, while the average duration of target segments and untrimmed videos are 8.1 and 30.6 seconds, respectively.

TACoS^[126]. The TACoS dataset contains 18 200 language queries paired with cooking videos, with an average length of 10.5 words per query. On average, each video lasts 4.78 minutes, while target segments are 27.9 seconds long.

MAD^[127]. The MAD dataset is collected for hours-long video temporal grounding, comprising 384 600 queries with an average of 12.7 words per query. It includes 1 207.3 hours of video, where each untrimmed video lasts 110.8 minutes on average, and target segments are 4.1 seconds long.

5.6.2 Evaluation Metrics

Intersection over Union (IoU). IoU@*m* is adopted as a metric under the recall rate of top-1 prediction. A predicted proposal is correct only when the IoU be-

tween the prediction and ground-truth is greater than the predefined IoU threshold m . The threshold m is usually set to $\{0.3, 0.5, 0.7\}$.

5.6.3 Performance Comparison

We list the performance of existing methods on the two most widely used datasets, Charades-STA and ActivityNet Captions. We list the performance of fully-supervised VTG models in Table 12. For the fully-supervised methods, one-stage methods not only are more efficient than two-stage methods, but also have obvious performance advantages over two-stage ones. For MLLM-based methods, their unique advantage lies in their strong zero-shot generalization capa-

bility. Therefore, in Table 13, we list the zero-shot performance of these methods. The MLLM-based methods benefit from the powerful multimodal understanding and reasoning capabilities of MLLM, showing significant advantages in both fully supervised and zero-shot scenarios.

The performance of weakly-supervised VTG models is listed in Table 14. Recently, weakly-supervised methods have advanced significantly, which have comparative performance with early supervised methods, especially when the IoU threshold is set to 0.3.

6 Applications

Beyond the fundamental tasks discussed in previ-

Table 12. Performance Comparison of Video Temporal Grounding Models on Charades-STA and ActivityNet Captions Dataset in the Fully-Supervised Setting

Category	Model	Source	Charades-STA			ActivityNet Captions	
			IoU@0.3	IoU@0.5	IoU@0.7	IoU@0.5	IoU@0.7
Two-stage	SAP ^[89]	AAAI 2019	–	27.42	13.36	–	–
	QSPN ^[90]	AAAI 2019	54.70	35.60	15.80	27.70	13.60
One-stage	UniVTG ^[91]	ICCV 2023	72.63	60.19	38.55	–	–
	UVCOM ^[92]	CVPR 2024	–	59.25	36.64	–	–
MLLM-based	TimeChat ^[93]	CVPR 2024	–	46.70	23.70	–	–
	TimeSuite ^[95]	ICLR 2025	79.40	67.10	43.00	–	–
	TRACE ^[96]	ICLR 2025	–	61.70	41.40	–	–
	Time-R1 ^[98]	arXiv 2025	82.80	72.20	50.10	55.60	34.00

Note: Higher metric values indicate better.

Table 13. Zero-Shot Performance Comparison of MLLM-Based Methods on Charades-STA and ActivityNet Captions Datasets

Category	Model	Source	Charades-STA			ActivityNet Captions		
			IoU@0.3	IoU@0.5	IoU@0.7	IoU@0.3	IoU@0.5	IoU@0.7
Pretrain	TimeChat ^[93]	CVPR 2024	–	32.2	13.4	–	–	–
	VTimeLLM ^[94]	CVPR 2024	51.0	27.5	11.4	44.0	27.8	14.3
	TimeSuite ^[95]	ICLR 2025	69.9	48.7	24.0	–	–	–
	TRACE ^[96]	ICLR 2025	–	40.3	19.4	–	37.7	24.0
	LLaVA-ST ^[97]	CVPR 2025	63.1	44.8	23.4	–	–	–
	Time-R1 ^[98]	arXiv 2025	78.1	60.8	35.3	–	–	–
Training-free	ChatVTG ^[112]	CVPR 2024	52.7	33.0	15.9	40.7	22.5	9.4
	VTG-GPT ^[113]	Applied Sciences 2024	59.5	43.7	25.9	47.1	28.3	12.8
	TFVTG ^[114]	ECCV 2024	67.0	50.0	24.3	49.3	27.0	13.4

Note: Higher metric values indicate better.

Table 14. Performance Comparison of Video Temporal Grounding Models on Charades-STA and ActivityNet Captions Dataset in the Weakly-Supervised Setting

Model	Source	Charades-STA			ActivityNet Captions	
		IoU@0.3	IoU@0.5	IoU@0.7	IoU@0.3	IoU@0.5
CNM ^[99]	AAAI 2022	60.04	35.15	14.95	55.68	33.33
CPL ^[100]	CVPR 2022	66.40	49.24	22.39	55.73	31.37
SCANet ^[101]	ICCV 2023	68.04	50.85	24.07	56.07	31.52
PPS ^[102]	AAAI 2024	69.06	51.49	26.16	59.29	31.25

Note: Higher metric values indicate better.

ous sections, cross-modal retrieval techniques have found extensive practical applications across diverse real-world domains. While the full spectrum of potential applications is broad, we focus here on several representative and impactful scenarios that illustrate the transformative potential of cross-modal retrieval in solving complex multimedia problems and enhancing user experiences.

Digital Commerce. Digital commerce platforms often face scenarios where customers want to find products through natural language descriptions or voice queries^[140]. Cross-modal retrieval systems address this need by enabling text-to-image or text-to-video search, delivering visually relevant product content that matches user descriptions. Such enhanced search capabilities allow customers to discover products more intuitively and make better-informed purchase decisions, improving both product exploration and comparison.

Identity-Based Person Search. Cross-modal retrieval supports identity-based person search by allowing users to locate specific individuals using natural language descriptions of appearance, clothing, or behavior^[141]. For example, queries like “person wearing a red jacket and blue jeans” or “tall person with a backpack” can retrieve relevant individuals from large-scale surveillance networks, crowd monitoring systems, or security databases. This functionality facilitates efficient person tracking and identification in safety and security applications.

Smart Content Creation and Editing. Cross-modal retrieval also enhances creative workflows by enabling content creators to search for multimedia assets using natural language descriptions. Queries such as “sunset over mountains with warm lighting” can retrieve relevant stock photos or video clips. This capability streamlines content creation for video editors, graphic designers, and social media creators, reducing the time spent manually browsing large media libraries and supporting more efficient and intuitive creative decision-making.

7 Future Directions

In this section, we discuss potential directions for future research. By adopting a broad perspective, insights can be drawn through comparisons both within and across tasks.

Open-Domain Subcategory-Level Fine-Grained Retrieval. Current subcategory-level fine-grained re-

trieval methods are primarily limited to closed-set scenarios due to the scarcity of fine-grained annotations. Mainstream benchmarks, such as CUB-200-2011^[130] with 200 bird species and Stanford Cars^[131] with 196 car model types, only cover a small fraction of the real world, which hinders the practical application of fine-grained cross-modal retrieval (FCMR) models. While vision-language pre-training (VLP) models such as CLIP and SigLIP have achieved remarkable success in open-domain coarse-grained retrieval through large-scale weakly-supervised pre-training, they lack the discriminative power required to distinguish between subtle fine-grained categories. Consequently, a pivotal future direction lies in breaking the dependency on limited fine-grained labels. The challenge is to develop paradigms for semantic FCMR that can scale to arbitrary fine-grained categories, bridging the gap between large-scale coarse-grained pre-training and specialized fine-grained semantic understanding.

Unified Multimodal Retrieval Foundation Models. The trajectory of AI is shifting towards general-purpose multimodal foundation models, such as GPT-5 and Gemini, which integrate understanding, retrieval, and generation across text, image, audio, and video. In this context, retrieval is no longer an isolated task but a core capability, raising the question of whether specialized retrieval models will be superseded by unified multimodal retrieval foundation models^[142, 143]. A central challenge lies in constructing a unified representation space that accommodates heterogeneous modalities and varying granularities, while ensuring both efficiency and universality. Promising directions include hybrid paradigms that combine a universal backbone with domain-specific adapters. Ultimately, a unified multimodal retrieval foundation model should be designed to support cross-modal retrieval from any modality.

Data-Centric Revolution. As model architectures reach maturity and incremental performance gains from novel designs diminish, the field is increasingly embracing a data-centric paradigm, recognizing that the quality, scale, and diversity of training data are the primary determinants of model capability. This shift is exemplified by initiatives like DataComp^[135], where researchers improve performance not by changing the model but by curating superior datasets from massive, noisy pools. In cross-modal retrieval, this opens several compelling directions. First, developing robust data filtering and curation strategies is critical: how can we effectively remove noise, mitigate bias,

and maximize semantic diversity across modalities and granularities? Second, designing dataset-level evaluation metrics that capture not just size but conceptual richness, the distribution of hard negatives, and the coverage of rare concepts will guide more principled data construction. Finally, synthetic data generation, such as using LLMs to produce high-quality captions, descriptions, or cross-modal alignments, offers a powerful means to supplement scarce or noisy annotations. Advancing cross-modal retrieval increasingly depends on innovations in dataset design and engineering, complementing architectural progress to fully leverage the potential of unified multimodal models.

User-Centric Retrieval. Future cross-modal retrieval systems should emphasize user-centered design, incorporating interactive mechanisms that adapt to diverse user goals and preferences^[144]. Key research directions include developing feedback-driven learning approaches that improve retrieval accuracy based on user responses, implementing query modification techniques that dynamically refine searches according to user intent, and creating intuitive interfaces with effective visualization tools. These interactive systems should facilitate query clarification, preference specification, and result exploration to enhance user understanding and decision-making in retrieval tasks.

Emerging Modalities. Beyond the traditional scope of images, video, and text, the integration of emerging modalities represents a frontier for cross-modal retrieval, effectively bridging the gap between digital content and physical sensory data. 1) In embodied AI and robotics, agents can align visual observations with motor commands or sound sources to interact with the physical world effectively. For example, a robot can identify the initiator of a command via a sound signal and can use 3D grounding to recognize and locate the object to interact with. 2) In autonomous driving, cross-modal retrieval facilitates sensor fusion and data mining by aligning 2D camera images with 3D LiDAR point clouds or radar signals. This enables the retrieval of historical driving scenarios with similar geometric or kinematic signatures. 3) In the medical domain, the focus extends to aligning 1D bio-signals (e.g., EEG, ECG) with medical imaging (e.g., MRI) to retrieve similar pathological patterns across modalities. Unified frameworks capable of handling signals from these diverse, asynchronous emerging modalities will be essential for next-generation multimodal intelligence.

8 Conclusions

Fine-grained cross-modal retrieval (FCMR) is a fundamental but challenging task, which brings the single-modal research and application to a new area. This survey presented a systematic overview of the recent works of CMR from coarse-grained and fine-grained perspectives, focusing on four representative tasks: image-text retrieval, video-text retrieval, image grounding, and video temporal grounding. We first categorized and summarized existing methods for these four tasks. Then, frequently used datasets and evaluation metrics were introduced to enable performance comparisons among representative methods. Furthermore, we discussed the applications and potential future directions in CMR. We expect this survey could not only provide a valuable overview of CMR, but also act as a catalyst for future research.

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